

**“EVALUATION OF PURCHASE AND CREDIT TERMS IN
HINDUSTAN AERONAUTICS LIMITED (HAL’S) DEFENCE
CONTRACTS”**



HINDUSTAN AERONAUTICS LIMITED

**Master Thesis submitted in partial fulfillment of the requirements for the award of the
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ABSTRACT

The research "Evaluation of Purchase and Credit Terms in Hindustan Aeronautics Limited (HAL)'s Defence Contracts" delves into how credit and purchase terms affect supplier performance and procurement efficiency in a leading aerospace and defence organization in India. The paper examines in detail whether contractual terms on credit period, discounts for early payment, interest on late payment, delivery period, and payment terms have a significant impact on operational and financial outcomes in the HAL procurement system.

The research makes use of a descriptive research design that combines qualitative and quantitative methods. A vast array of secondary data was extracted from 400 supplier records, HAL's annual reports, and open defence procurement documents. Besides, SPSS was used for performing various statistical analyses such as correlation and regression tests to measure and determine the strength of the relationships between Credit Utilization, Average Payment Delay, Actual Delivery, and Delay variables.

It was established through the research that credit terms affect supplier performance very slightly as can be discerned from the weak correlations and the low coefficient of determination ($R^2 = 0.016$). In the same way, purchase terms showed little influence on procurement efficiency with Delivery Period being the only variable to some extent Actual Delivery but not significantly affecting overall delays ($R^2 = 0.003$). The results from this research embody those operational practices, supplier reliability, process management are the key factors that are instrumental in procurement effectiveness apart from formal credit or purchase clauses.

The research rounds up with a verdict that organized credit and purchase schemes provide financial discipline but are not potent enough on their own to drive supplier efficiency. HAL is advised to take on a holistic procurement strategy which focuses on, among other things, on-line supplier monitoring, workflow automation, data-driven decision-making, and strategic supplier engagement. The latter being essential for both transparency and performance. Besides, supplier relations can be strengthened through continuous feedback and the use of performance-based incentives which can further enhance accountability and contract fulfillment.

CHAPTER – 01

INTRODUCTION

1.1 OVERVIEW OF THE TOPIC

This study, covering a wide scope, deals with the financial and supply chain management aspects in the defense sector. While the study is general in nature, it tends to delve deeper into the research of purchasing policies, procurement systems, and credit terms that are structured, analyzed, and evaluated in big public sector undertakings (PSUs) such as Hindustan Aeronautics Limited (HAL). Procurement and credit management are the foundation of industry operations, ensuring resources are acquired in an efficient manner and financial obligations are met without interruption. In defence-related operations, procurement and credit management take on a greater strategic importance because they directly impact national security, operational readiness, and the timeliness of defence-related projects. Hindustan Aeronautics Limited (HAL), a premier public sector undertaking in India, illustrates the importance of procurement and credit management functions in an environment of significant regulation and specialization.

HAL's primary functions include the design, manufacture, maintenance, and overhaul of aircraft, helicopters, engines and avionics systems for the Indian Armed Forces. These processes require a dependable and consistent supply of raw materials and parts, and specialized materials. Any delays in procurement due to credit management issues, or procurement process issues, degrades the project's schedule, and to some extent an associated or implicit delivery, which could even jeopardize the operational readiness of the defence forces. HAL's procurement system is therefore more than a basic business process, it is a strategic enabler of the country's defence industrial base.

The procurement function in HAL is both complex and highly structured. It consists of identifying a need based on production schedules, sourcing potential suppliers, assessing potential suppliers' technical and financial capabilities, contracting negotiations, and ensuring timely delivery of materials in a structured procurement process. HAL has obligations to comply with federal legislation for procurement institutions and higher reporting requirements than other industrial procurement processes. Consequently, the entire procurement process in HAL is administered in detail by the rules and requirements set by the public sector undertaking and the firm ultimately

secured in a contract for a functional component, parts or services of the end product for the operations outlined in the contract process.

Moreover, by examining purchasing and credit terms at HAL, one can understand how the company manages financial discipline as well as operational efficiency in a large and complicated organization in the defense sector. The procurement procedure at HAL is based on the principles of openness, fairness, and responsibility. Each process, from invitation to bid to the final vendor selection, is carried out according to the established government regulations and public procurement policies. This not only assures that the company is getting the highest quality of materials at fair prices, but also that there is an equal playing field among suppliers. Such a methodical process contributes to the elimination of the risk of delays, lowering the risk of finance, and creating the conditions for close and long-lasting business relationships with vendors.

It is just as necessary for the proper running of HAL that credit terms be managed. Good credit management provides the business with the opportunity to balance its payments, keep up with a good rapport with customers, and guarantee that their cash flow will be used for other operations. By determining definite credit periods and assessing the financial reliability of suppliers, HAL makes sure that payments are made in the right way and on time. This not only prevents the company from facing financial problems but also ensures that suppliers remain incentivized to deliver materials on time.

The accomplishment of HAL in handling the purchasing and credit terms in an effective manner is a clear indication that well-planned financial systems can back up the production of sizeable volumes of the country's requirements for national defense. The firm's dedication to moral business conduct, openness, and the economical use of resources clearly illustrates how it continues to be one of the most reliable Indian defense agencies. The seamless synchronicity of the Finance and Operations divisions at HAL is a prime example for other PSUs which aspire to reinforce their purchasing and credit systems.

1.2 THEORETICAL BACKGROUND OF THE STUDY

In order to analyze HAL's assessment of purchasing and credit terms, it is important to highlight some relative ideas and frameworks of procurement and financial management. Procurement theory focuses on efficiency, fairness, and strategic sourcing, primarily influenced by the Five R's - Right Quality, Right Quantity, Right Price, Right Time & Place, and Right Source.

1.2.1 Procurement Management Theories

The Five R's provide a useful lens through which to evaluate the procurement management approach HAL employs when using defence contracts.

- Right Quality - ensuring that materials or services conform to predefined technical specifications (i.e., performance characteristics). The Purchase Manual outlines a robust quality-first approach with detailed quality checks, certifications, and inspections by designated inspection authorities.
- Right Quantity - balancing the need to procure in many small orders and want to procure in larger quantities to reduce overhead costs. HAL requires prospective vendors to submit detailed Material Purchase Requests (MPRs) documenting the quantities they are ordering.
- Right Price - not necessarily the lowest, but the prices that are fair and justifiable, based on a market assessment of the product, the vendor's capabilities, and sustainability over the contract term. HAL completes the evaluation of its contractual commitment requirement with a detailed comparative statement prior to enacting the contract being presented.
- Right Time & Place - ordering in time to be consistent with production schedules. Disruption of the flight-line implementation schedule can delay aircraft assembly completely, therefore scheduling is very much a priority for HAL.
- Right Source - selection of a reliable vendor, based on financial stability, technical capability and historical performance. HAL requires vendors to be listed in the Vendor Directory of the purchase manual and updates the directory every two years.

1.2.2 Theories Related to Financial Management

When viewed from a financial perspective, credit terms fall within working capital management. Credit terms address the timing of payments; the timing of payments can occur from prepayment, paid on milestones or post-delivery. Theories of liquidity management suggest,

Shorter credit periods allow HAL to generate cash flow, but may adversely impact their vendor relationships. Extended credit periods may be beneficial to the vendor but lead HAL to incur additional financial liabilities.

HAL mitigates this duality through Performance Bank Guarantees, Security Deposits and associated payments to milestones.

1.2.3 Contractual Theory

Contractual theory elucidates how risk-sharing provisions, warranties, dispute resolution, and penalties are embedded in contracts. HAL's contracts usually contain these provisions for:

- Earnest Money Deposits (EMD) and Security Deposits to ensure vendor commitment.
- Performance Bank Guarantees to protect against non-performance.
- Arbitration as a dispute mechanism.

The theoretical grounding demonstrates that HAL's procurement includes risk sharing, efficiency, financial prudence, and legal protections—but often at the expense of increased procedural complexity.

1.2.4 Agency Theory

Agency Theory clarifies the connection between one party, the principal (HAL), and the other party, the agents (vendors/suppliers). In cases of defence contracts, HAL relies on vendors to provide important parts on time and at the quality cost.

However, there are always the potential risks of vendor opportunism, delayed delivery, or cost over runs. To mitigate this, HAL utilizes Earnest Money Deposits, Performance Bank Guarantees, inspection protocols, and milestone payments. These contractual mechanisms serve to realign the vendors' incentives and goals closer to HAL's incentives and goals—in other words, changing the divergence of utility that HAL and the vendor/supplier have--when they potentially diverge. This serves to alleviate information asymmetry and protect HAL from vendor/supplier non-performance.

1.2.5 Transaction Cost Economics (TCE)

Transaction Cost Economics, championed by Oliver Williamson, indicates that organizations will try to structure their transactions to minimize the possible costs associated with negotiation, monitoring and enforcement. In HAL's case, working with multiple defence vendors could result in high transaction costs for HAL due to the costs associated with tender processes, quality assessments, and compliances. To resolve this situation, HAL specifies procedures for sourcing and monitoring vendors through its Directory of Vendors and in future procurement and credit arrangements by means of long-term contracts, and through standardization within the Purchase Manual. These actions reduce uncertainty by developing trust within a known vendor directory, and improves efficiency in sourcing, procurement, and credit arrangements. By minimizing transaction costs, HAL can allocate resources to their core component of aircraft manufacturing while still maintaining transparency and accountability.

1.3 EXPLANATION FOR RELATED CONCEPTS OF SELECTED RESEARCH

All academic research involves a clear grasp of the significant ideas that underpin the research. In this study, which examines the purchase and credit terms of Hindustan Aeronautics Limited (HAL) in its defence contracting, some key interrelated concepts are relevant. These concepts help conceptually ground the study and provide a way of thinking about procurement and financial practice in an organization the size and significance of HAL.

1.3.1 Procurement

Procurement essentially represents an ordered process in which an organization can locate, source, and purchase the goods and services necessary to successfully carry out their functions. In the defence industry, procurement is much more important than in ‘normal’ industry because the quality, reliability, and timeliness of delivered goods directly impact national security and defence readiness. A private firm, for example, may focus much of its efforts in procurement to achieve the best price point. However, a public sector enterprise (such as HAL) is required to function using the Government of India’s Defence Procurement Procedures, making procurement a strategic and highly regulated process.

1.3.2 Purchase Policies

Purchase policies constitute the main core of HAL's purchasing framework by setting out a clear and uniform method of buying the necessary materials, components and services for the defense operations. These policies go through the whole procurement cycle, besides vendor registration and pre-qualification, they include tendering procedures, bid evaluation, quality inspections, contract finalization and post-delivery verification, and so on.

HAL's Purchase Manual, the reference document for these policies, is the instrument that guarantees transparency, fairness and accountability, as it brings every purchase into line with government regulations, defense procurement procedures, and organizational objectives. The policies further provide for the prevention of risk, the ethical compliance, and the economical use of finances through the imposition of performance bank guarantees, security deposits and milestone-based payments.

Knowing well that the vendor performance and the past interactions are the prerequisites for successful projects, HAL, by means of a well-maintained Vendor Directory, is not only keeping track of these but also making the selection of suppliers for critical projects a strategic decision. The regular audits, internal reviews, and compliance with statutory norms are the instruments which assure a consistent enforcement of policies that purchase in turn, help to keep down disputes, shorten procurement of delays, and grant to acquire of high-quality materials at competitive prices. Besides that, these policies are the foundation of long-term, trust-based partnerships with vendors that result in innovation, dependability and operational continuity in the highly specialized aerospace and defense sector.

1.3.3 Credit Terms

Credit terms refer to the payment conditions that are agreed upon between a buyer and a supplier. Credit terms and conditions of payment may be categorized as advance payments; milestone payments, or payments on a post-delivery basis. For example, in the case of HAL, credit terms become important as HAL balances its cash flow requirements with its long-term, cooperative vendor relations. Specifically, shorter credit cycles enhance HAL's liquidity but can exert pressure on vendors; longer credit cycles help vendors maintain their operations but increase HAL's financial commitments. In a commitment to balance these competing interests, HAL utilizes mechanisms such as performance bank guarantees and security deposits, which allow HAL to extend credit terms without putting itself entirely at risk.

1.3.4 Working Capital and Liquidity

Handling working capital is a crucial factor that is intertwined with the credit terms a company decides to give its suppliers and customers. Good working capital management at HAL means that the company can very well handle its short-term debts and also be able to finance its ongoing operations simultaneously. Liquidity is extremely important particularly in a heavy manufacturing company like HAL, where late deliveries of components or materials can directly lead to aircraft or helicopter assembly coming to a standstill. In order to keep the plant running, HAL is taking care of the cash inflows and outflows in such a way that payment to suppliers is done on time while making the best use of the credit periods granted.

1.3.5 Vendor Relationship Management

The defense manufacturing industry is a sector in which the different needs of the industry must be provided for by the stable network of suppliers. The efficient organization of vendor relationships is the main factor that guarantees the smooth flow of operations and the accomplishment of the project goals. Vendor relationship management at HAL does not stop at the stages of simple transactional interactions; it is a process of constructing long-lasting, strategic partnerships in which both parties have common goals, share risks, and benefits.

1.3.6 Risk Management in Contracts

Risk management in contracts forms an essential part of HAL's procurement and operational framework as defense contracts are complicated in nature and consist of high-value, specialized components. These kinds of contracts are exposed to a wide range of risks such as delays in delivery, cost overruns, poor quality, technological failures, and disputes over payments or contractual terms. HAL anticipates these risks and deals with them by means of legal, financial, and operational safeguards.

1.3.7 Regulatory and Institutional Framework

In conclusion, examine the context in which HAL operates are an important aspect of any study on procurement and credit terms in HAL.

HAL is a public sector undertaking and, as such, is constrained by government rules, audit requirements and defence procurement rules and regulations. The Defence Procurement Procedure (DPP) and Defence Acquisition Procedure (DAP) regulate procurement processes and specify, for example, that HAL must offer three vendors by way of competitive bidding.

The Purchase Manual, meanwhile, translates the aforementioned defence procurement compliance rules into HAL's organizational procedures. For HAL, the institutional framework constrains procurement and credit processes and procedures to ensure that the procurement and credit processes are transparent and competitive while also maintaining a certain degree of accountability to the Canadian public and complying with the larger objectives of national defence.

The study was designed to cover major concepts such as procurement, purchasing policies, credit terms, working capital, vendor relationships, risk management, and regulatory frameworks, all of which have a significant impact on HAL's defense contracts.

HAL's procurement is just not about buying the parts only; it is a story of getting secure, high-quality components from local as well as global suppliers while following government standards that are very strict and quite difficult to meet. The Ministry of Defence and the national objectives such as "Make in India" are the primary factors molding its purchasing policies that demand the organization to be transparent, efficient, and future-oriented. Since defense programs are very long and extremely costly, credit terms are very significant as well, as it is imperative for HAL to strike a balance between the payments to vendors and the long credit periods from its customers.

Studying all these factors together gives the research both the thickness and the background, which not only helps in understanding how HAL's systems work but also where there would be the possibility of smooth and strong changes.

CHAPTER – 02

ORGANIZATIONAL PROFILE

2.1 INTRODUCTION TO THE ORGANIZATION



(Figure 2.1 Founder of (HAL))

Hindustan Aeronautics Limited (HAL) is one of India's leading aerospace and defence organizations. Founded in 1940, courtesy Walchand Hirachand's vision to establish aircraft production in India, HAL began as a small workshop focused on the minor repairs and assembly of airplanes. Today, HAL has transitioned into a large organization that designs, develops, and maintains fighter aircraft, helicopters, engines, and avionics systems.

HAL operates under the purview of the Ministry of Defence of India and is a critical component to ensure aircraft for India's armed forces are dependable and modern. HAL is not exclusively dedicated to aircraft production; they have made considerable strides to independently conduct research and develop technologies for the future such as the Light Combat Aircraft (Tejas) and several generations of helicopters.

Beyond defence, HAL has also significantly provided for the civil aviation sector while conducting maintenance, training, and technical assistance. HAL operates numerous factories throughout the nation and is highly skilled workforce of engineers and technicians. Balancing HAL's extensive growth from a small workshop to an aerospace leader, reflects how determination, vision, and work ethic, have transitioned a small company to a leader across the aerospace market in India.

HAL has systematically established self-reliance in defence and **aerospace**, and it continues to grow even today.

2.1.1 History of the organization



(Figure 2.2 Logo of (HAL))

Hindustan Aeronautics Limited's history and evolution, exclusively, is linked with the development of the Aeronautical industry in India for more than 85 years.

The Company which had its origin as Hindustan Aircraft Limited was established on 23 Dec 1940, at Bangalore by Shri Walchand Hirachand, a man of vision, with the joint collaboration of the then Government of Mysore to manufacture a/c in India. The Government of India became a shareholder in the company with effect from March 1941 and took over management of the company in 1942. The Company commenced its business of manufacturing the Harlow Trainer, the Curtiss Hawk Fighter and the Vultee Bomber aircraft, in collaboration with the Inter-Continental Aircraft Company of USA.

In January 1951, Hindustan Aircraft Limited came under the administrative control of the Ministry of Defence, Government of India. Hindustan Aircraft Limited manufactured aircraft and engines of foreign design under licence; e.g., the Prentice, Vampire, and Gnat. It also designed and developed aircraft systems indigenously. In August 1951, the very first flight of the company's

HT-2 Trainer system designed and produced under the proficient leadership of Dr. V.M. Ghatge took place.

The company manufactured and supplied more than 150 HT-2 Trainers to the Indian Air Force and other customers. As the design capability gradually built up, the company successfully designed and developed four other aircraft including it: two-seat Pushpak, a system suitable for flying clubs, Krishak for Air Observatory Post (AOP) role, HF-24 Jet Fighter (Marut), and HJT-16 Basic Jet Trainer (Kiran).

In August 1963, the Government of India established Aeronautics India Limited (AIL) as a wholly-owned government company to manufacture MiG-21 aircraft under license. Factories were established at Nasik (Maharashtra) and Koraput (Orissa). In June 1964, the Aircraft Manufacturing Depot set up in 1960 in Kanpur as an Air Force unit for producing the airframe of the HS-748 transport aircraft was transferred to AIL. Shortly thereafter, it was decided to amalgamate Hindustan Aircraft Limited with AIL to conserve resources in the aviation sector where we don't have a deep pool of technical talent, and to enable the planning and coordination of activities covering all aircraft production entities, in an overall efficient and economical manner.

The merger of Hindustan Aircraft Limited and Aeronautics India Limited took place on October 1st, 1964, through an Amalgamation Order by the Government of India, and the name of the merged company became Hindustan Aeronautics Limited (HAL), and its principal activities included design, development, manufacture, repair, and overhaul of aircraft, helicopters, engines, and related systems, such as avionics, instruments, accessories, etc. In 1970, a separate division was set up only for the manufacture of 'Chetak' and 'Cheetah' helicopters in Bangalore under a license agreement with M/s SNIAS, France, and a new division was set up to manufacture aircraft instruments and accessories at Lucknow. License agreements were entered into with M/s Dunlop, U.K. for Wheels and Brakes; Dowty for undertakings and hydraulic equipment; Normal Air Garret for cabin air pressurization and air-conditioning equipment; Smiths of U.K.; SFENA and SFIM of France for panel instruments and gyros; Martin Baker U.K. for ejection seats; and Lucas for engine fuel systems, all related to Marut, Kiran, Ajeet, Chetak, Cheetah, and Jaguar. A similar type of arrangement was agreed upon with USSR authorities for their aircraft MiG-21 series for the manufacture of accessories.

In 1971 the Avionics Design Bureau at Hyderabad was established for the development and production of IFF, UHF, HF, Radio components, Radio Altimeter, Ground Radars, and other technologies.

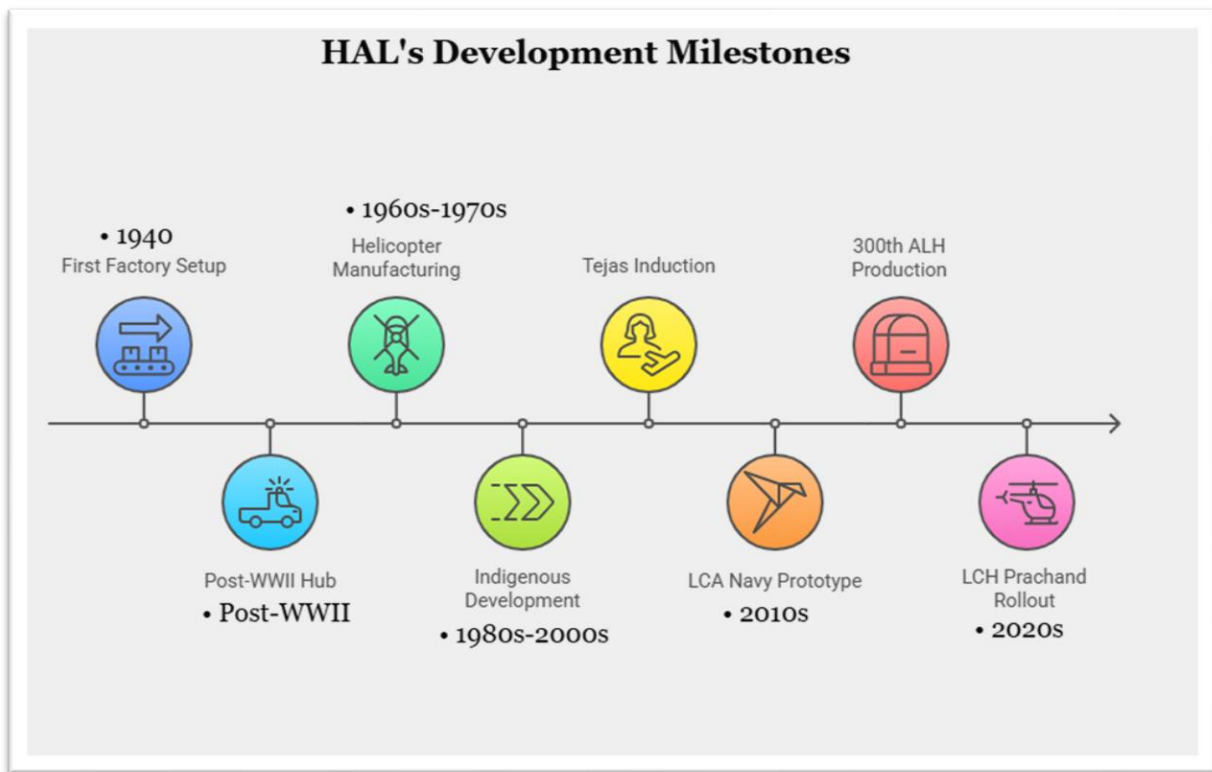
In 1973, a Design unit was established at Lucknow for the design and development of accessories which included undercarriage and hydraulic systems, air conditioning and pressurization systems, fuel control/gauging systems, generator control and protection units, static inverters, etc.

In 1979, after pursuing a licence agreement with British Aerospace, the Company commenced manufacture of the 'Jaguar' aircraft and with Rolls Royce-Turbomeca for Adour engines. The Company also entered into various licence agreements with other firms for the manufacture of avionics and accessories.

In 1982, the Company entered into an agreement with the USSR and started production of the Swing-wing MiG-27M aircraft as a follow-on project for MiG-21 BIS at the Nasik Division of the Company. HAL is engaged and is contributing to the space programs of the country. An Aerospace division was formed in 1988. HAL is currently supplying structures to the Division for aerospace launch vehicles, and satellite requirements of ISRO. It has also established infrastructure to execute the assembly of strap-on L-40 stage booster. HAL has supplied structures to ISRO for GSLV Mk.III, Mars Mission, and Human crew module. HAL is establishing dedicated facility for manufacture of cryogenic engines. In May 2000, a separate profit centre was set up to provide Airport related services in order to synergize the operations of HAL Bangalore Airport. The main goal for the creation of this Airport Service Centre was to reorganize the existing resources to provide specialized services specifically tailored to the service market relating to airlines operations and to commercially exploit the current resources of the Company at Bengaluru.

2.1.2 Key Milestone & Influences

➤ Milestone:



(Figure 2.3 HAL's Development Milestones)

➤ Influence:

- **Self-Reliance:** HAL's commitment to self-reliance in aerospace and defense technology is rock-solid; HAL promotes the indigenous ability to design, develop and produce.
- **"Make in India":** HAL is in the vanguard of the 'Make in India' initiative to foster cutting-edge aerospace technology in India.
- **International Partnerships:** HAL's collaborations with elite global players, (for example Rolls-Royce, General Electric and Safran), have greatly enhanced HAL's technological

capability and international standing.

- **Diversification:** HAL has diversified its products, moving from aircraft to a variety of helicopters, engines, avionics systems, UAVs and now civil aircraft.
- **Technological Advancement:** HAL is committed to research and development to advance indigenous technology; HAL is heavily engaged with progressing capability in fighter aircraft and helicopters.
- **Infrastructure Development:** The establishment of specialized wings and facilities, such as a new Aircraft Composite Division and a new Division at Koraput, which are conducive to enhancing HAL's manufacturing and MRO capabilities.

2.1.3 Key aspects of the Indian aerospace market.

The most important features of the Indian aerospace market, where Hindustan Aeronautics Limited (HAL) is at the core, are robust government support and indigenization programs, a fast-expanding domestic civil aviation industry, HAL's proven expertise in design, development, and manufacture of a wide variety of indigenous aircraft and systems, emphasis on R&D and technology development, high offsets and collaborations with foreign players, and growing private sector involvement fueled by initiatives such as "Make in India".

The following aspects highlight the importance of the Indian aerospace market with a focus on HAL's contributions:

- **Indigenous Design & Manufacturing:** HAL possesses full-fledged design and development capabilities for several types of aircraft, which include the first indigenously conceived and built jet fighter, the HF-24 Marut and HF-17 Tejas light combat fighter aircraft.

- **Diverse Product Portfolio:** HAL also manufactures a wide range of products, including fighter jets, helicopters, jet engine, marine gas turbine engine and avionics.
- **Support & Upgrades:** HAL provide essential support and maintenance services, including the refurbishment and upgrade for the existing Indian military aircraft.
- **Strong R&D Capabilities:** The Company has considerably enhanced its research and development (R&D) centers to support its growth and improve India's capacities in advanced aerospace technology.
- **Technology Harvesting & Innovation:** HAL's effort to use and innovate on existing global technologies, such as jet engine technology and AI-based technology, is meant to lessen reliance on foreign sources and advance domestic innovation.
- **Fostering an Ecosystem:** HAL's large-scale programs, such as the licensed production of the Su-30MKI, have developed a strong supply chain, involved many Indian suppliers and jobs, and added to the Indian aerospace industrial basis.

2.2 COMPANY PROFILE

2.2.1 Nature of business

Hindustan Aeronautics Limited (HAL) is India's leading aerospace and defense company under the Ministry of Defence. HAL's business is mainly related to the design, development, manufacture, repair, overhaul, upgrading, and servicing of a diverse array of aircrafts, helicopters, engines, avionics, and aerospace-related products.

- **Core Manufacturing & Production:** HAL is engaged in the indigenous production of fighter aircrafts, helicopters, transport aircrafts, jet engines and allied systems. HAL is an essential partner of India's defence services with state-of-the-art aeronautical military equipment.

- **Research & Development (R&D):** HAL has a strong commitment to R&D, working independently or in collaboration with domestic and overseas industry or military, to create indigenous platforms such as the Light Combat Aircraft (LCA-Tejas), Advanced Light Helicopter (ALH-Dhruv) and Light Combat Helicopter (LCH).

HAL are seeking to innovate in aerospace technologies and to ensure future self-reliance (to align with the "Atmanirbhar Bharat" and "Make in India").

- **Maintenance, Repair & Overhaul (MRO):** HAL provides support for the maintenance, repair, and overhaul of aircrafts and helicopters in co-operation with the Indian Air Force or the Indian Navy, Army, Coast Guard or from civil operators. This ensures operational readiness and lifecycle support for defence assets.
- **Licensing and Collaborations:** HAL engages in the production of aircraft and engines under license through partnerships with global leaders such as Boeing, Airbus, Rolls-Royce, and Sukhoi. HAL collaborates with aerospace companies internationally in order to facilitate technology transfer and joint development of aircraft and aerospace technologies.

2.2.2 Vision, Mission

Vision of HAL

- HAL wishes to be recognized as a prominent name in the international aerospace industry.
- It needs to make India capable and strong in defense and aviation technology.
- Company is keen on making its products safe, reliable, and of quality.
- Development through continuous improvement and innovation is a significant part of its vision.

Mission of HAL

- To design, produce, and maintain aircraft, helicopters, engines, and allied systems for the nation.
- To provide contemporary and affordable solutions for both defense and civilian aviation requirements.
- To continue developing research and new technology so India is not as reliant on imports.
- To collaborate with other Indian and foreign organizations and also engage in exports.

2.2.3 Evaluation & Growth of the HAL

Evaluation of HAL

Evaluation in an organization simply means assessing how well it is achieving its objectives. Evaluation is a process which is structured to establish performance, validity, and effectiveness. The purpose is to see whether the organization and service provided meets its objectives, and identify strengths and weaknesses, inform improvements, and provide accountability to stakeholders such as; shareholders, employees, government, and society. Evaluation is broader than a financial audit, which only looks at the accounts. Evaluation incorporates: finance, day-to-day operations, strategy, employees, governance, and utilizes both quantitative data and qualitative feedback. As a Maharatna Public Sector Undertaking (PSU) in the aerospace and defence sectors, Hindustan Aeronautics Limited (HAL) is subject to scrutiny via various evaluative mechanisms.

- Financial evaluation: HAL is subjected to study through examination of its annual balance sheet, profit and loss account, and cash flow so as to review growth, profitability and sustainability. It has been established that key ratios and the size of HAL's order book indicates the strength of their future business opportunities. The ratio examining cost efficiency and contract closures also reveals the level at which HAL has successfully resolved inappropriate expenditures during contract fulfillment.

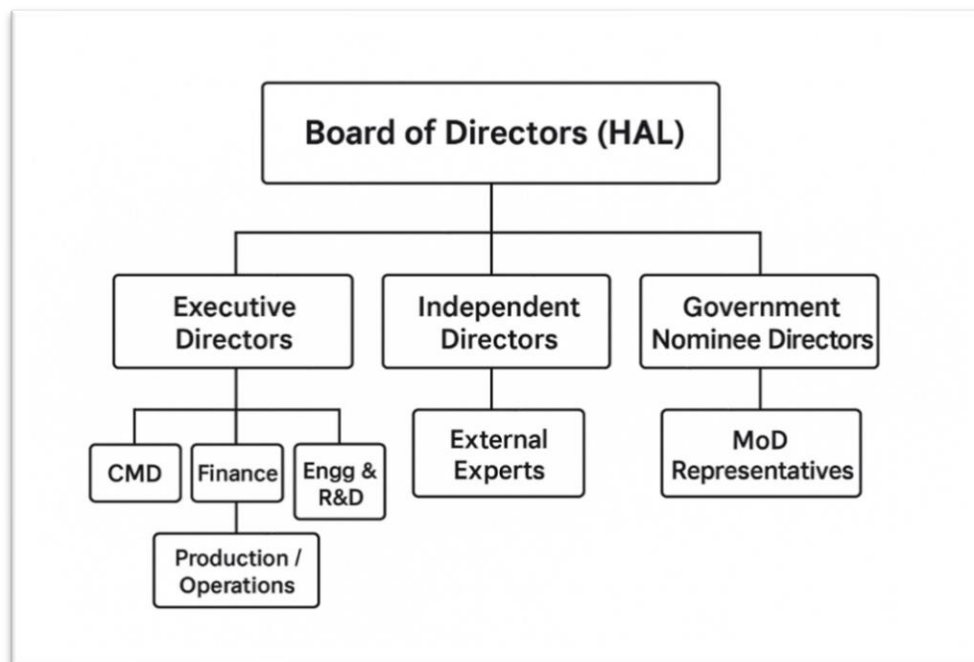
- Strategic & operational evaluation: HAL's performance against its corporate plans, work in progress on projects such as the Tejas fighter and Prachand helicopter, and examination of its use of production capacity, all provide measures of achievement against corporate goals in a timely manner and of an appropriate quality.
- Procurement & credit practices: The evaluation system used by HAL also assesses transparency in vendor selection, fairness in tendering, a systematic approach to supply inspection for all supplies and materials, and transparency in payments.
- Human resource evaluation: HAL performs and reviews evaluations of employee performance, training assessment and effectiveness and workplace relations, so as to ensure it has talented and engaged employees and an appropriate level of workforce stability across HAL complex sites.

Growth of HAL

- Early Years (1940s–1960s): HAL began in 1940 by Walchand Hirachand. The government took over later. Initially, they simply assembled the British planes for the most part. They collaborated with Hindustan Aircraft and Bristol Aeroplane Company as well. During the 1960s, they assisted in producing MiG-21 jets with the Soviet Union. This was a big thing for India, you know.
- Expansion (1970s–1990s): After that, they began producing engines, helicopters, and other components. They developed India's first jet, HF-24 Marut. They also acquired the skills to repair and maintain various aircraft on behalf of the army, navy, and air force. This benefited the nation considerably.
- Producing Own Aircraft (2000s–2010s): During the 2000s, HAL concentrated on producing their own aircraft. They produced Tejas fighter aircraft, Dhruv helicopters, and LCH. They also conducted a lot of R&D and set up design centers all over India. It was a major leap towards becoming self-sufficient.

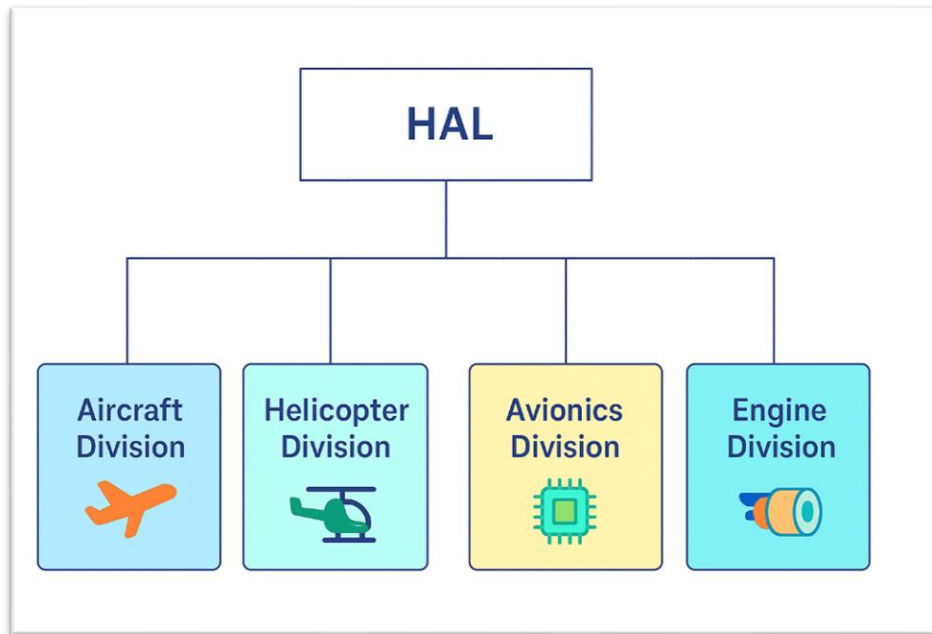
- Recent Times (2010s–present): HAL grew into a large government corporation and began earning more money. They also sell helicopters and spares to some other nations. They cooperated with organizations such as Boeing, Airbus, Rolls-Royce, and Sukhoi in order to learn new technology and produce better aircraft. That assisted them quite well, which is good.
- Future Plans: HAL is producing new aircraft like AMCA and Twin-Engine Deck Based Fighter for the Navy. They are also developing drones, small civilian planes, and space missions with ISRO. They wish to export more planes overseas. HAL began small but now it is among India's most significant defense firms, which is really awesome.

2.2.4 Boards of Directors



(Figure 2.4 Board of Directors of HAL)

2.2.5 Divisions



(Figure 2.5 Division's Charts)

2.2.6 Bangalore Complex:

- Aircraft Division
- Engine Division
- Industrial & Marine Gas Turbine Division
- Overhaul Division
- Aerospace Division
- Foundry & Forge Division
- Airport Service Center

- LCA/LSP
- IJT/LSP
- Central Materials and Process Laboratory

2.2.7 The Helicopter Complex in Bangalore Includes:

- Helicopter Division
- Helicopter MRO Division
- Rotary Wing Research and Design Center
- Composite Manufacturing

2.2.8 MIG Complex:

- Sukhoi Engine Division Koraput
- Aircraft Manufacturing Division Nasik
- Overhaul Division Nasik

2.2.9 Achievements & Awards

Hindustan Aeronautics Limited (HAL) has garnered notable recognition, such as the SCOPE Eminence Award for Institutional Excellence in August 2025 due to its exemplary performance and service to the aerospace and defence industry, and the National Award for Innovative Training Practices in 2019 as a result of its contributions to human resource development. HAL is also credited with the development of cutting-edge indigenous platforms such as the LCA Tejas and India's defense manufacturing self-reliance, as explained on its official website.

Key Achievements of HAL:

- **Design, Development & Production:** HAL has been a key contributor to the design, development, and production of different aircraft, helicopters, engines, and avionics. These achievements were not only beneficial for the Indian defense industry, but also to India as a rising force in world defense technology.
- **Indigenization:** One of HAL's strongest immediate capabilities has been the development of indigenously designed and developed platforms. The LCA Tejas, Dhruv helicopters, and other advanced systems demonstrate HAL's ability to minimize dependence on foreign defense suppliers, as well as encourage a self-sustaining manufacturing base in India.
- **Development of Advanced Technologies:** Apart from aircraft and helicopter development, HAL is constantly seeking advances of new technologies. HAL has developed a number of significant products such as advanced display units, mission computers, and automatic flight control systems. The existence of these technologies enables India to avail modern defense technologies designed in India.
- **Aircraft & Helicopter MRO:** Apart from production, HAL is also involved in maintaining, repairing, and overhauling (MRO) aircraft and helicopters and their systems. HAL's total MRO task in support of aircraft and helicopters ensures that the machine remains reliable, economical, and utilizable over the working life.

Recent HAL Awards



(Figure 2.6 HAL Awards Distribution)

- SCOPE Eminence Award for Institutional Excellence (2025): HAL received this award based on Maharatna/Navratna, as a recognition of the Company's overall achievements and its various contributions in providing products & services to National aerospace and defence.
- Excellent PSU of the Year (2024): The award recognize HAL as one of the high achievers in terms of Public Sector Undertaking and confirming company's leadership and cognitions with regard to the industry.
- National Award for Innovative Training Practices (2019): Indian Society for Training & Development (ISTD) awarded HAL for innovative and pragmatic training programmes for the key role in strengthening and enhancing company's employees', skills and capabilities.

- India Today CPSE Excellence Award (2014): HAL was recognized with this award under Best Global Presence based on HAL's growing visibility and recognition on the international stage.
- India Pride Award (2013-12): HAL was presented with this award for its consistent and outstanding performance and in recognition of its contribution to strengthening the national defence.

2.2.7 HAL Today

Key Developments at HAL

- Tejas Mk1A Fighter Aircraft:
 - i. First Flight in October 2025: HAL's Nashik unit intends to deliver the first Tejas Mk1A combat aircraft in the third week of October. This is a significant achievement in India's defense manufacturing capabilities in times of self-reliance.
 - ii. Big Order Booked: The Indian Defence Ministry has signed a ₹62,370 crore contract with HAL for the procurement of 97 Tejas Mk1A jets, which includes the delivery of 68 single-seater aircraft and 29 twin-seat aircraft and deliveries are expected to commence in 2027-28.
- HTSE-1200 Turboshift Engine:
 - i. Development Update: HAL is progressing the development of the HTSE-1200 turboshift engine for 3.5-ton single-engine and 5-8-ton twin-engine class helicopters. The initial batch of engines will be delivered by mid-2025.
- Restructuring for Enhanced Efficiency:
 - i. Government intending changes: To deal with its ₹2.7 lakh crore order book and shorten

lead times of deliveries, the Indian government is looking to restructure HAL.

- ii. A consulting firm has been hired to evaluate HAL's current structure and operations to improve efforts and speed up delivery to the military.
- International Expansion Initiatives:
 - i. Global Cooperation: HAL is looking at ways to broaden its global reach, including partnerships and cooperation initiatives, that will expand its international presence and strengthen the 'Make in India' agenda.
- Financial Projections
 - i. Stock Market Performance: HAL's stock has gained by about 15% during 2025, demonstrating investor faith in the wake of major defense orders and strategic moves. Analysts estimate a possible rally to ₹6,220 on the back of improved revenue and improved indigenization.

2.2.8 Financial Performance of HAL

Hindustan Aeronautics Limited (HAL) continues to show robust financial health in FY 2024–25, demonstrating stable growth in revenue, profitability, and strategic investments. The performance of the company showcases its significant contribution to India's aerospace and defense industries.

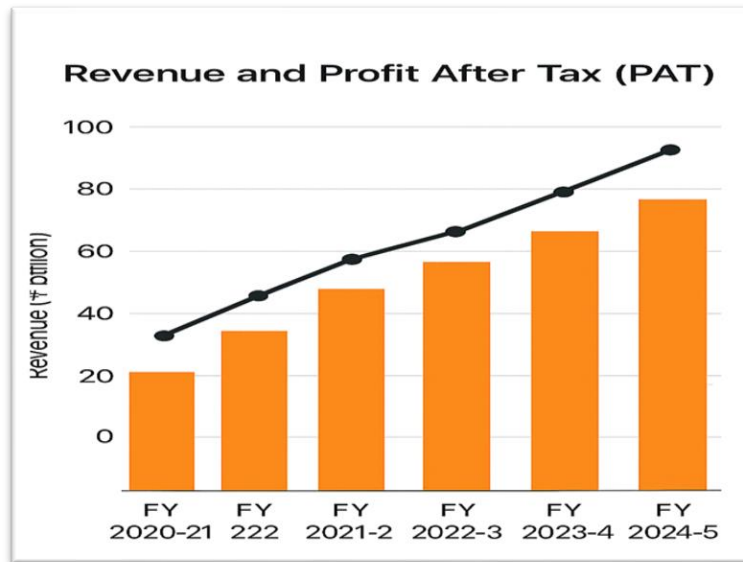
HAL Financial Performance Overview (FY 2024–25):

- **Revenue and Profit Growth**

For the years 2024 and 2025, Hindustan Aeronautics Limited (HAL) completed a financial performance that stands out among its previous efforts. The main revenue of the company went higher than last year's ₹28,100 crore by 7%, reaching ₹30,105 crore.

The growth of HAL has been largely due to the continuation of defence orders, among which Tejas, helicopters, and the rapidly growing HAL's maintenance, repair, and overhaul (MRO) services sector stand out. In terms of profitability, HAL's Profit Before Tax (PBT)

was ₹10,820 crore, thus, increasing from ₹10,199 crore reached in 2023–24. Also, its Profit After Tax (PAT) went up to ₹8,317 crore, witnessing a 10% annual growth. It is very important to note that such figures not only point to the operational strength and expediency of HAL but also demonstrate the company's capacity to keep up with profits as well as allocating resources for future capabilities that is a strategy which guarantees its continued position as the India's aerospace and defence industry is the backbone.



(Figure 2.7 Revenue & PAT Graph)

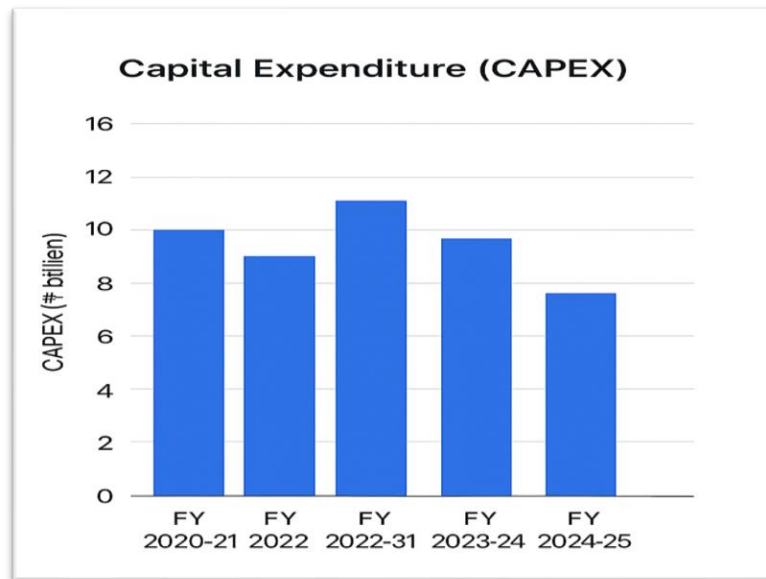
➤ Earnings Per Share (EPS)

For the financial year 2024–25, HAL has revealed a Basic and Diluted EPS of ₹124.27, pointing to the excellent dividends the company has given its shareholders. These numbers represent the company's substantial earnings and the successful control of expenses across the various business segments.

➤ Capital Expenditure (CAPEX)

HAL went ahead with the strategic investments in its facilities to the tune of ₹2026 crore, just a bit lower than ₹2168 crore in FY 2023–24. It was essentially these investments that initiated the expansion of helicopter manufacturing capabilities, the modernization of production facilities, and the upgrade of MRO services for the users of both the domestic

and international markets.



(Figure 2.8 CAPEX Graph)

➤ **Operational Highlights**

HAL's helicopter and aircraft production lines are functioning smoothly at their full capacity and are able to meet delivery schedules for the Indian Armed Forces as planned. HAL has developed its maintenance, repair, and overhaul (MRO) department, which guarantees that defense aircraft and helicopters have good availability and are in good operational conditions. Measures for self-reliance like the local production of aircraft engines and high-tech systems have not only lowered the dependence on foreign imports but also made it possible for the country to save money and become technologically self-reliant.

2.2.9 CSR Activities

1. Introduction

Hindustan Aeronautics Limited (HAL), a Navratna Central Public Sector Enterprise under the Ministry of Defence, Government of India, is a leader in Corporate Social Responsibility (CSR). The firm understands that business success over a period of time is still dependent on factors other than financial performance, such as sustainable development and social well-being. The company's CSR plan at HAL is envisioned to create a positive impact on the society through the promotion of integral socio-economic development, betterment of the community, and conservation of the environment.

2. CSR Vision and Objectives

Vision: "We aim to be a major contributor to the combined social and economic evolution of the neighborhoods where HAL is functional, and moreover, to care for the environment and ensure the perpetuation of nature."

Objectives:

- Firstly, through the interventions of our corporate social responsibility, we shall promote education, healthcare, and skill development.
- Secondly, we endeavor to empower the local communities by engaging with them in activities exclusively targeted at the improvement of their daily lives.
- Thirdly, apart from the social aspects, CSR will remain committed to the total sustainable use of natural resources and environmental care through co-profit projects.

3. Major Focus Areas of HAL's CSR Activities

a) Education and Skill Development

Building and refurbishing school buildings, classrooms, and laboratory facilities in the rural and semi-urban localities. Offering educational scholarships and materials to the economically weak children of the society. Government and private school digital learning initiatives and provision of smart classrooms are also a part of the strategy to revive the education system. Youth counseling centers for the development of employment-related skills and trades mainly in aerospace, machining, and technical fields could be created through the assistance of the HAL CSR activities. Industrial Training Institutes (ITIs) and private partnerships may also take action to promote the modernization of training and infrastructures that can offer the learning of the new-to-the-industry curriculum and the skill requirement of a particular profession.

b) Healthcare and Sanitation

Implementation of medical camps in remote areas that would deliver free health check-ups... and necessary medicines to the unserved people. Covering expenses of and managing primary healthcare groups and mobile medical units around the HAL locations with the revenue raised from CSR activities. Among the programs of Swachh Bharat Abhiyan that are supported by the company include the building... of toilets in schools and areas accessible to the public. There are also hygiene, nutrition, maternity health, and preventive healthcare awareness programs conducted by the staff. One more area in which HAL has been actively engaged is the support of COVID-19 relief initiatives through the supply of oxygen plants, PPE kits, and vaccination campaigns.

c) Environment and Sustainability

Implementation of extensive tree planting programs, and ecology creation, not least through... green belts that surround the locations of HAL. The company encourages the usage of renewable energy and has fitted the solar power system to the building to which it is used by the office of the company. Waste management actions have been financially supported and physically executed by the... company, which include waste segregation and recycling. Water from rain would be saved in containers and small lakes and also pools in close vicinity to HAL towns would be brought back

to life by these kinds of projects. At the same time the company is also very committed to the implementation of awareness campaigns, that would help the people understand not only the issue of climate change but also the biodiversity and different methods through which the environment can be conserved.

d) Rural Development and Community Welfare

Building a road network, community halls, and domestic water supply in the local villages... nearby the company facilities. In collaboration with local government districts the company has begun the implementation of the rural electrification projects. Women's Self-Help Groups (SHGs) will receive the company's support through Confidence Training. Program and micro-finance initiatives. Programs of Sustainable Livelihood in the field of tailoring, handicrafts, and agriculture training will gradually be introduced to the members of society. One of the ways through which the youth can be motivated is the development of sports facilities

4. Flagship CSR Initiatives

Some significant CSR initiatives implemented by HAL are:

- HAL CSR School at Ganjam (Odisha) – ensuring quality education for the children of the economically weaker section of society.
- HAL Hospital in Bengaluru – offering healthcare services not only to the employees but also to the neighboring community.
- Skill Development Programs in partnership with NSDC (National Skill Development Corporation).
- Village adoption program under HAL divisions for rural development in a comprehensive manner.
- Donation to PM CARES Fund and relief activities in different parts of India.

2.2. Sustainability

1. Introduction

As foremost aeronautics and defence public sector enterprise in India, Hindustan Aeronautics Limited (HAL), is committed to leading by example as it infuses sustainable development into its comprehensive operations. HAL is of the view that the survival of the business in the long run is dependent on a combination of financial results and environmental intimacy, employee wellness, and social commitment. The company's sustainability objectives center around maintaining a stable equilibrium between the advancement of technology, saving the earth, and the growth of society.

2. Sustainability Philosophy

HAL's sustainability philosophy is based on the company's adherence to the triple bottom line concept — People, Planet, and Profit.

- People: Ensuring good living conditions for employees, safety, and community development.
- Planet: Trying to reduce the environmental damage by using resources judiciously.
- Profit: Keeping the wheel of the economy turning by coming up with new ideas and being efficient in business.

The company aims at sustainable development through the synchronization of its business goals with the United Nations Sustainable Development Goals (SDGs), in particular, those concerning industry innovation, clean energy, quality education, and responsible consumption.

3. Environmental Sustainability Initiatives

The main activities are:

- Energy Conservation: Use of energy-saving and advanced technology-based equipment and LED lighting throughout the unit of the factory. Solar power plants have been installed

in major divisions such as Bengaluru, Nashik, and Koraput. It is the organization's constant effort to find energy-saving potential through energy audits and awareness programs.

- **Water Management:** Setting up of rainwater harvesting is done in all the major institutions. Among the measures being implemented are recycling of wastewater and building of treatment plants that uses water for gardening and utilities. Water is better used through the drip irrigation system in green areas.
- **Waste Management:** Practice of 3R principles – Reduce, Reuse, and Recycle – is implemented. Separation of different types of wastes, including hazardous and non-hazardous waste, for disposal in a proper way. The measures being implemented to reduce the production of chemical and industrial waste during the process of manufacturing.
- **Green Initiatives:** Planting trees in parks and streets in the vicinity of HAL facilities to increase the green cover. OL maintenance of eco-friendly campuses and green belts. Participation in World Environment Day and related sustainability campaigns.

4. Economic and Operational Sustainability

- HAL is committed to maintaining its financial position of power while at the same time following socially responsible, honest, and transparent business practices.
- By continuous upgrading of production plants, the company is also successful in cutting its production costs and saving the environment through the efficient use of resources.
- The firm also reduces its dependence on foreign suppliers and, at the same time, facilitates the domestication of defence production by investing in the in-house research and development (R&D) department. With the help of actively promoting indigenous manufacturing and supplier development, the company supports the two government flagship schemes “Make in India” and Atmanirbhar Bharat.”

5. Social Sustainability

HAL's sustainability framework covers social responsibility and human capital development as well. Essential components of the program are:

- Guaranteeing safe and healthy work environments in all departments/divisions.
- Offering skill development and training programs for employees and apprentices.
- Encouraging gender equality and diversity in the organization.
- Implementing CSR projects in education, health, sanitation, and rural development (linked with CSR section).
- Helping the local communities through job creation and providing the means for socially desirable growth.

6. Governance and Ethical Practices

- HAL is governed by a well-functioning corporative governance system that, among other things, ensures the respect of the principles of transparency, accountability, and ethical conduct.
- The company is also committed to environmental and occupational safety management by following the standards for these areas set by ISO 14001 & ISO 45001.
- The company has also established sustainability committees at both corporate and divisional levels to supervise the implementation and the conformity to the set standards.
- Engaging regularly with stakeholders on sustainability issues through its annual report and disclosures.

Conclusion

HAL's commitment to sustainability is deeply embedded in its business model and not barely a separate function. The company's eco-friendly social and community-friendly economic initiatives have made it a viable and green company of the future. Following the lead of the government and the United Nations Sustainable Development Goals, HAL is making a significant contribution to the aerospace sector's green transformation in India.

2.3 PRODUCT AND SERVICE PROFILE

- Aircraft
- Helicopters
- Aero Engines
- Avionics and Systems
- Aerospace Structures and Components
- Maintenance Repair & Overhaul (MRO)
- Technical Support & Upgrades
- Design & Development
- Training & Simulation
- Supply Chain & Logistics Support

Additional Product and Service Details

- Unmanned Aerial Vehicles (UAVs) / Drones
- Space Systems & Components
- Specialized Defense Equipment
- Research & Development Services
- Indigenous Manufacturing Initiatives
- Export and International Services



HAWK – (Figure 2.9 HAL Hawk Advanced Jet Trainer Aircraft)



ALH- (Figure 2.10: HAL ALH Aircraft)



RUDRA- (Figure 2.11: HAL Rudra)



MIRAGE – (Figure 2.12: Mirage)



GROUND RADAR - (Figure 2.13: Ground Radar System used by HAL)



HYDRAULIC PUMP (Figure 2.14: Hydraulic Pump used in HAL Aircraft Systems)



FLIGHT DATA RECORDER (Figure 2.15: Flight Data Recorder)

2.3.1 Products

1. HAL Light Combat Aircraft (LCA) – Tejas



(Figure 2.16: HAL Light Combat Aircraft (LCA) – Tejas)

The LCA Tejas is a 4.5 generation, all-weather, and multi-role fighter aircraft. It has been designed as a multi-role aircraft that can easily take up offensive air support, aerial dogfighting as well as the ground attack role. Besides, it is also a Ground - Maritime styled Operations fighter. The different variants of aircraft under production and development are:

- Air Force single-seat fighter
- Navy single-seat fighter
- Air Force twin-seater trainer aircraft
- Navy twin-seater trainer version

Moreover, LCA Mk1A is the most sophisticated model of the LCA Tejas. It comes with AESA Radar, an EW suite made up of radar warning and self-protection jamming, Digital Map Generator (DMG), Smart Multi-function Displays (SMFD), Combined Interrogator and Transponder (CIT), Advanced Radio Altimeter, and other advance features.

2. HAL IJT (Intermediate Jet Trainer)



(Figure 2.17: HAL IJT)

The Intermediate Jet Trainer (IJT) christened as HJT-36 is indigenously designed and developed by HAL to replace ageing fleet of KIRAN Jet Trainer aircraft in service with Indian Air Force for Stage II training of its pilots. IJT will be fitted with AL-55i Jet engines produced in house at Engine Division Koraput. IJT incorporates the simplicity necessary for ease of conversion from Basic Piston Trainer and the sophistication required for quick conversion to the complexities of an Advanced Jet Trainer.

- Two-seat, intermediate jet trainer aircraft
- Purpose: The aircraft is designed for transition training from basic trainers to advanced fighter jets as well as training pilots to high-performance fighter aircraft.
- Characteristics: The plane was designed to have a clean and aerodynamic body which was optimized for training and handling the aircraft in a quick way.
- Powerplant: One turbofan engine (generally HAL-developed or made under license).

3. HAL Dhruv (Advanced Light Helicopter – ALH)



(Figure 2.18: HAL Dhruv)

The HAL Dhruv is just a helicopter of multiple abilities, having two engines, and being suitable for different types of work like transport, observation, rescue, and medevac operations. It is designed from the hard composite materials and fitted with the advanced avionics; thus, it can function effectively in a rough atmosphere. The armed models can be employing an array comprising rockets and other armaments. Besides assembling the Dhruv, HAL also takes care of its servicing and overhauling under defense contracts and thus one of the company's major revenue contributors.

- Twin-engine – A safe and stable aircraft, when only one engine is operating.
- Multi-purpose – Besides transport, the aircraft can be used for observation, rescue and medevac.
- Sturdy airframe – Made of lightweight composite for difficult areas.
- Sophisticated avionics – Accurate navigation and handling.
- Capacity for high altitudes – Works well in mountainous areas.
- Can be armed – It is possible to equip the aircraft with rockets, missiles, and guns.
- MRO and upgrades – Maintenance and system updates are done by HAL.

4. HAL Do-228 (Civil Aircraft)



(Figure 2.19: HAL Do-228)

HAL Do-228 is an aircraft with two engines, which, among other things, has the capability to take off and land on short runways and is used for both the civil and military sectors. With its dependability and tough construction as a selling point, in fact, it can perform all the way from short and unprepared airstrips, which makes it perfect for various operations like transport of the region, surveillance, and passenger services. Apart from building the Do-228, HAL through agreements, also provides repairs, upgrading's, and technical assistance, thereby being a major contributor to the company's earnings from both the civil and defense sectors.

- Twin engines – A safe and reliable choice for any flight.
- STOL capability – The plane is able to both take off and land in a short and/or unprepared runway.
- Multimodal usage – The aircraft is the primary choice for the transport of the passenger, the freight, and the observer.
- Heavy-duty airframe – Built to survive the hard times.
- State-of-the-art avionics – Equipped with the latest in navigation and communications technology.

5. GTRE GTX-35VS Kaveri



(Figure 2.20: GTRE Kaveri Engine)

The GTRE GTX-35VS Kaveri is an Indian native turbofan engine created by the Gas Turbine Research Establishment (GTRE) specifically for the HAL Tejas plane. The engine was conceived to provide outstanding performance and to be efficient, it co-opts the use of innovative materials and state-of-the-art design to stand the tough requirements of the fighter aircraft. The collaboration between HAL and GTRE allows the process, the testing, and the servicing of the engine, thus, it is an important element of India's effort to achieve technological self-sufficiency in the defense sector.

FEATURES:

- **High Thrust Performance:** The engine can create a maximum thrust of 81 kN along with an afterburner giving the pilot the possibility of superior fighter jet performance.
- **Indigenous Technology:** The Pyrojet MK-1 was made by GTRE (DRDO) with the use of cutting-edge materials and the FADEC system for the efficiency and control of the engine.
- **Modular Design:** Its modular design makes it convenient to be taken care of through regular maintenance and repair work.

6. Safran Ardiden 1H1 (Shakti)



(Figure 2.21: Shakti Engine)

The Safran Ardiden 1H1, famously referred to as the Shakti engine, is a turboprop engine that comes from the collaboration of HAL and Safran. The engine unit is the power source for the HAL Dhruv helicopter, and the Light Combat Helicopter (LCH). It is a high-performance engine concept for "hot and high" operational modes and also comes with a state-of-the-art dual-channel FADEC system for accurate and effortless management. HAL is responsible for the manufacture, quality control and repair of the engine, which is thus an extremely important part of the Indian helicopter program and defense tenders.

FEATURES:

- **High-Altitude Performance:** The aircraft has the capability to operate in "hot and high" conditions, which is extreme temperatures and high-altitude missions, hence the power is stable and reliable.
- **Advanced FADEC System:** The engine is set up with a Fully Digital Engine Control (FADEC) system that is electronically controlled, it consists of two independent channels for safety, and it operates efficiently and automatically.
- **Joint Development & Production:** HAL and Safran have jointly developed the engine, while HAL has taken up the responsibility of the manufacturing, quality control, and maintenance, therefore, the national defense self-reliance is empowered.

2.3.2 Services

- Maintenance, Repair, and Overhaul (MRO) – HAL offers end-to-end MRO support for both military and civilian aircraft. This covers scheduled maintenance, unscheduled overhauls, and repair of components to keep the aircraft safe and airworthy.
- Technical Support & Upgrades – HAL provides technical support, systems upgrade, and life extension schemes for its aircraft and helicopters. This involves avionics upgrade, performance improvement, and upgrading existing fleets to keep up with changing operational needs.
- Design and Development Services - HAL is more than just a manufacturer, it also makes a significant contribution to the design and development of airplanes, helicopters, engines, and other systems that go with them. Apart from the design, they also work on prototype development and perform testing in association with the Indian Defence Research & Development Organization (DRDO). Besides the change of the standard specifications, HAL is also involved in the change of the feature requirements of the defense to make the aircraft more suitable for electronic warfare or for using as a UAV.
- Avionics and Electronics Integration - HAL is the one who provides the avionics system integration service that involves the implementation of navigation, radar, communication, and electronic warfare systems, besides aircraft becoming compatible with the latest combat or operational requirements. In addition, HAL manages the changes in software and the integration of the mission-specific electronic system as well.
- Component Manufacturing & Assembly - HAL produces the most critical components for the aeronautical sector such as the landing gear, turbine blades, the fuselage, aero-structures and carries on assembly operations for the military as well as the civil platforms. This physical integration promotes the maintenance of trust, the upholding of standards and compliance with aerospace regulations.

2.4 MCKINSEY'S 7-S FRAMEWORK



(Figure 2.22: McKinsey's 7-S Framework)

1. Strategy:

- **HAL's Strategic Goals:** HAL has its main concentration on the production of the defense and aerospace sectors that are independent, carbon-supporting the country's security, and being a part of the "Make in India" initiative. The company aims to achieve through research, development, and manufacturing an India-based military and civil aviation technology besides the Tejas Light Combat Aircraft (LCA) and the Dhruv Advanced Light Helicopter.

- **Expansion and Diversification:** HAL is progressively expanding its boundaries by collaborating with international partners and exploring different fields such as unmanned aerial vehicles, avionics, and engine production. The company is accomplishing a double victory with the diversification of its business, as it goes beyond traditional markets while also making a significant contribution to the country's 'Make in India' mission in the aerospace and defense sector.

2. Structure:

- **Organizational Setup:** HAL is a public sector unit under the Ministry of Defense, India. It is an enterprise with diverse divisions and units spread throughout the country to facilitate different production, research, and service activities.
- **Decentralized Operations:** The company's designs its operations to accommodate the existence of various hubs for the production, MRO (maintenance, repair, and overhaul), and R&D. In this way, the aerospace company's structure promotes and allows the concerned divisions to concentrate on such product categories as helicopters, fighter aircraft avionics, etc.

3. Systems:

- **Operational Systems:** HAL uses systems to manage production planning, supply chain, and quality that meet extremely high standards set for the defense and aerospace sectors. The company has processes that are strong enough to manage inventories, project timelines, and compliances.
- **Innovation and R&D Systems:** HAL Inc. is heavily investing in R&D systems, by collaborating with overseas and local organizations to keep up with the latest technology and also maintaining design and development capabilities for native products.

4. Shared Values

- **Core Values and Vision:** The HAL values are based on the satisfaction of customers, comprehensive quality commitment, cost-effectiveness, innovation, honesty, and team spirit. Its vision is to lead the world over in the aerospace and defense sector, giving the most advanced technology and services.
- **National Mission Alignment:** Being a public-sector undertaking, HAL's principles are consistent with the defense objectives of the country India, tasks of reliance and toughness in defense production being underlined by it.

5 Style

- **Leadership Style:** The management style of HAL is a combination of both a top-down and a participative approach, with a close alignment to government policies and strategy objectives on the one hand and fostering inter-departmental cooperation and sharing of ideas on the other hand.
- **Decision-Making:** HAL is known to make decisions based on the nature of its public-sector operations while strictly following the regulations, wisely allocating the budget, and promoting openness in the management processes.

6. Staff

- **Employee Expertise:** The employees of HAL consist of the best engineers, highly trained technicians, and experienced managers in the aerospace and defense sectors. The company encourages talent growth through comprehensive training activities and collaboration with educational and research institutions.

- **Employee Engagement:** HAL focuses on the commitment of employees, a corporate climate of mutual confidence, and continuous personal growth, which is necessary for maintaining the particularized technical abilities in aerospace and defense industries.

7. Skills

- **Technical and Operational Expertise:** The core competencies of HAL are the production of the aircraft, R&D in avionics and engines, the provision of maintenance services, and systems integration. With these capabilities, HAL is able to execute turnkey projects such as the Tejas LCA and Sukhoi Su-30MKI.
- **Versatility and Creativity:** HAL is developing the necessary skills in new domains like UAVs and advanced avionics for maintaining its position in the rapidly changing sector. HAL's partnership with the foreign companies facilitates skill acquisition and the easy transfer of knowledge.

CHAPTER – 03

REVIEW OF LITERATURE AND RESEARCH DESIGN

3.1 REVIEW OF LITERATURE AND GAPS

1. Tuti, Edi Krisharyanto, and Fani Martiawan Kumara Putra (2025). “Binding Force of Purchase Order as an Agreement”. *International Journal of Social Science and Human Research*, Tuti, Krisharyanto, in their paper demonstrate how purchase orders may be taken beyond just routine paperwork. If they define the fundamental contract elements – offer, acceptance, intention, and consideration, then these become binding in law. Such a step enables companies like Hindustan Aeronautics Limited (HAL) to not only secure their legal rights but also augment the supplier relationship, cut down on conflicts and raise the operational efficiency level of the organization. Besides, the research also points out that a pledge to arrangement creates the conditions for trust, loyalty and superior performance among the employees.

2. Imran Khan (2021). “Multiple-Criteria Decision Making (MCDM) Methods for Aerospace & Defence Industry Supplier Selection”. *International Journal of Social Science and Human Research*, the paper presented by Imran Khan in 2021 implements multiple-criteria decision-making (MCDM) methods for the supplier selection at Hindustan Aeronautics Limited (HAL). The methods aim at evaluating the suppliers on several aspects including cost, quality, delivery, innovation, and financial stability. The study indicates that congruent supplier selection along with a positive credit term can lead to higher procurement efficiency, risk reduction, and supply chain resilience being strengthened. Techniques such as AHP, TOPSIS, and SWARA enable HAL to come up with a supplier ranking who characterize both technical and strategic choices.

3. Venkateh Balasubramanian (2016). “Financial Ratio Analysis of Hindustan Aeronautics Limited”. *International Journal of Social Science and Human Research*, in his study, Balasubramanian (2016) unpacks the financial ratio analysis of Hindustan Aeronautics Limited (HAL) as a financial health assessment and procurement strategy evaluation. The research delves

into the key ratios associated with profitability, liquidity, efficiency, and solvency, thus explaining how HAL has been able to gain its profits and liquidity, while also witnessing the possibility of the company identifying the areas of efficiency and solvency where improvements could be made. The paper conveys that by knowing all these financial indicators, HAL can be led to the best of purchase, control of the circulating capital and even raise the operational efficiency. Besides which, the study also points out that being financially strong is not only a good support for the management of the suppliers which in turn provides for the stability of the organization and its long-term strategic planning.

4. Abhijeet Nehra (2021). “Changes in Defence Procurement of India”. International Journal of Social Science and Human Research, Nehra explores the reforms in India's defence procurement that were undertaken primarily to make the process efficient and to encourage 'Make in India'. The paper sheds light on problems faced like the slow administrative processes and reliance on foreign imports, and underlines the importance of indigenization and co-operation with the private sector. The study finds that the optimization of the procurement system leads to higher military preparedness, promotion of the technology sector through research, and the achievement of national security goals by integrating defense strategies.

5. George Jumbe & Ravi Gor (2024). “Assessment of Credit Spread in Default Risk using Merton Model”. International Journal of Innovative Research in Technology, Merton model is applied to evaluate the credit spreads along with default risks of three companies: Hindustan Aeronautics Limited (HAL), Eicher Motors (EM), and Adani Enterprises (AE). The researchers assess the financial data from the balance sheets between March 2018 and March 2022 and decide to analyze the debt-to-asset leverage ratios of the companies to find out how much their credit spreads differ. The results show that HAL with a median leverage ratio of 0.67 is the company with a credit risk profile that lies between those of EM (low leverage ratio of 0.41) and AE (high leverage ratio of 0.96) with which it is compared.

6. Kanagavalli, G. (2019). “A Study on Working Capital Management of Hindustan Aeronautics Limited.” International Journal of Innovative Research, this paper investigates the current asset management at Hindustan Aeronautics Limited (HAL) and how it affects the purchasing process and credit terms. The report implements several financial analysis instruments such as liquidity ratios and funds flow statements to measure HAL's financial stability and

productive efficiency. Among the top issues addressed by the company is their plan for the concerted use of the current assets and current liabilities to secure production and distribution free of any bottlenecks. The study also points out that the need of the hour is to keep a right amount of working capital to not only support the company's activities but also to be able to fulfill its obligations. The effort is finished with a proposal on bettering working capital through effective management of the company's financial performance and engagement in operations.

7. Sharma, D. S., Patil, Y. D., & Kumar, R. (2021). “A Study on Inventory Management System: A Case Study of Hindustan Aeronautics Limited (HAL).” *International Journal of Innovative Research* 7(12), 45–52, this paper studies the inventory management (IM) practice in Hindustan Aeronautics Limited (HAL) and its influence on procurement efficiency and credit terms. The research applies various analytical tools to evaluate HAL's inventory control measures, disclosing the use of tactics such as stock level management, lead time, and supplier relations. The main results show that HAL's IM system is instrumental in the streamlining of the procurement process and the maintenance of good credit terms with suppliers.

The paper emphasizes the need for efficient IM to achieve operational performance and financial stability. Suggestions are made to further improve inventory control to increase the efficiency and cost-effectiveness of HAL operations.

8. Desai, Nayana (2024). “Skyward Bound: The HAL Odyssey – Innovations, Alliances and Trials.” *International Journal of Innovative Research*, the research unfolds how HAL's strategic co-operations and tweaking of contracts have deeply influenced its defense procurement architecture. By digging deeper into the company's partnership-building and negotiating credit terms, Desai portrays the way HAL marries technological innovation with money management. Besides, the article outlines the progress made in the implementation of contractual reforms and the role played by the involvement of suppliers in the eradication of the cause of the delay in procurements, the increase of the effectiveness of the defense sector in general.

9. Lo Nigro, Giovanna, Giovanni Favara, and Lorenzo Abbate (2024). “Supply Chain Finance: The Role of Credit Rating and Retailer Effort on Optimal Contracts.” *International Journal of Supply Chain and Financial Management (ResearchGate)*, In this research, the authors are looking to find out how credit ratings, payment terms, and retailer activities impact the

supply-chain finance contracts' design. They come up with a model that connects a company's financial solvency with its operational effectiveness, illustrating that well-functioning credit and purchase measures can go a long way in achieving better coordination between suppliers and prime contractors. The study also points out that a perfect match between credit terms and effort incentives can lead to lower default risks and less transaction inefficiency. These findings can be applied to companies like Hindustan Aeronautics Limited (HAL) which greatly depends on supplier collaboration and contract optimization to ensure the smooth running of production, maintain financial discipline, and have trust among the defense supply network.

10. Venkatesh, K. (2009). “Vendor Assessment and Procurement Decision Making through Data Mining in the Aviation Industry.” *International Journal of Advanced Engineering and Management Research (ResearchGate)*, In his 2009 work, Venkatesh deals with the use of data mining techniques to raise the level of vendor evaluation and the quality of procurement decisions in the aviation sector. The paper outlines the main points of an evaluation like payment history, credit terms, delivery timelines, and quality compliance, illustrating their impact on supplier selection and contract stability. The study underlines that the melding of analytical instruments into the procurement systems enables organizations to forecast vendor performance and streamline sourcing strategies. The results are very much applicable to the Hindustan Aeronautics Limited (HAL) as they present how credit and purchase terms can be used to deepen supplier relationships, lower the risk, and upgrade the operational efficiency of defense manufacturing.

11. Ramanathan, S., and Patel, Divya (2025). “Key Factors Driving the Potential Future Developments in Automated Manufacturing at HAL.” *Journal of Defence Production and Industrial Engineering (ResearchGate)*, Ramanathan and Patel (2025) have analyzed the changing landscape of the use of automated manufacturing processes in Hindustan Aeronautics Limited (HAL) with an extensive focus on how modernization by technology has changed the buying and supplier management practices. The article conveys that while HAL moves towards automation and digital assimilation, supplier ability and contract flexibility turn into the main features. The researchers have engaged in the discourse of the reforms in purchasing with a special emphasis on milestone-based payments, performance-linked contracts and vendor qualification processes under automation initiatives. These findings evoke the notion that future procurement

efficiency of HAL hinges on achieving a successful integration of automation with sound credit terms, supplier performance monitoring, and strategic vendor partnerships.

12. Jacob, George, and Bhatta, N. M. K. (2024). “The Conundrum of Self-Reliance in Defence Capability in the Indian Context: A Qualitative Study.” *Journal of Management and Entrepreneurship*, (ResearchGate), Jacob and Bhatta (2024) investigate the influence of procurement reforms, payment mechanisms, and self-reliance objectives on DPSUs such as HAL. The paper illustrates how contract arrangements and the anticipation of payments help harmonize procurement strategies with defence goals at the country level, thus offering HAL some pragmatic pointers for better supplier engagement and enhancement of operational capabilities.

13. Likhachev, Kirill A., and Makhlaiuk, Arthur N. (2024). “Major Factors that Impede Reforms in India’s Military-Industrial Complex (2014–2023).” *International Journal of Social Science and Human Research*, Likhachev and Makhlaiuk dissect the obstacles that are present in the institutions which choke the changes in India's military-industrial complex. They go deep into the analysis of bureaucratic inertia, lack of transparency, and outdated procurement processes as some of the main factors. Their research outlines the manner in which these limitations inject the inefficiency and the lack of competitiveness of the Defence Public Sector Undertakings (DPSUs), Hindustan Aeronautics Limited (HAL). Authors are calling for the implementation of broad policy reforms to facilitate procurement activities, strengthen supplier relations and invigorate defense sector with the new thinking.

14. Gupta, Ashish, Kumari, Puja, and Wadhwa, Jiya (2025). “An Analysis of Hindustan Aeronautics Limited (HAL): Becoming the 14th Maharatna Company in India.” *International Journal of Social Science and Human Research*, Gupta, Kumari, and Wadhwa (2025) review HAL's conferral of the Maharatna status, focusing its outstanding financial performance, and its independence to take strategic decisions. The report details the extent to which this rank provides HAL with increased working and fiscal autonomy, thus the possibility of carrying out large operations and taking up new projects. The writers speculate HAL's purchasing methods which now probably include matters like contract negotiations and payment policies, giving an overview of the changing nature of defence buying in the country.

15. Ballamudi, Satyanarayana K. (2024). “Analyzing Supply Chain Efficiency in HAL's Avionics Division Using SPSS: A Statistical Exploration.” *Journal of Aerospace Supply Chain Management (ResearchGate)*, Ballamudi (2024) makes use of SPSS statistical tools to measure the supply chain efficiency of Hindustan Aeronautics Limited (HAL) Avionics Division. The research work outlines essential sourcing factors of the organization like terms of payment, supplier lead times, and contract compliance that affect the functioning of the organization. Through the examination of these variables, the study offers the insights to the procurement strategies and payment methods that may positively impact supplier relationships and supply chain flow that is furthered by the HAL & apos; s avionics department.

16. Tiwari, R., & Singh, A. (2023). “Trade Credit Management Practices in Indian Defence Public Sector Undertakings.” *International Journal of Social Science and Human Research*, Tiwari and Singh (2023) have explored the usage of trade credit in Defence Public Sector Undertakings (DPSUs) which also comprises Hindustan Aeronautics Limited (HAL). This research throws up some of the important features of the investigation such as payment schedules, discussions on the credit period, and supplier financing, illustrating the benefits that go along with the organized credit terms in the procurement system, mitigating the risk of nonpayment, and also the facilitation of the supplier relationship.

17. Raghavan, S. (2022). “Supplier Relationship Management in Aerospace Manufacturing: Evidence from HAL.” *International Journal of Social Science and Human Research*, Raghavan (2022) examines the ways Industrial credit terms influence supplier performance in Hindustan Aeronautics Limited (HAL) through supplier relationship management with emphasis on how supplier performance is impacted by credit terms, milestone payments, and flexible contract structures. The research shows that the efficient handling of these purchasing and financial instruments contributes to the smooth operation of the supply chain, decrease in disputes, and deepening of supplier relations in the aerospace supply sector.

18. Choudhary, P., & Mehta, K. (2021). “Optimizing Payment Terms to Reduce Procurement Risk in Indian Defence Contracts.” *Journal of Defence Procurement and Financial Management (ResearchGate)*, Choudhary and Mehta (2021) in their paper looked into the effects of payment schedules, advance payments, and performance-linked credit terms on

procurement risk in Indian defence organizations like Hindustan Aeronautics Limited (HAL). Their research reveals that the use of a good credit and payment plan not only solves the problem of supplier reliability but also decreases the risk of default, and allows through procurement contracts, the implementation of risk-adjusted decision-making.

19. Srinivas, R. (2020). “Impact of Working Capital and Credit Policies on DPSU Procurement Performance.” *International Journal of Public Sector Finance and Management (ResearchGate)*, Srinivas (2020) studies the effects of the return on working capital, credit policies, cash cycle payable/receivable on the purchasing activities' efficiency in Hindustan Aeronautics Limited (HAL) and other DPSUs. The research points out that by adjusting credit terms with the suppliers' resources, companies can maximize the purchasing process time, minimize the occurrence of waiting for the delivery of goods, and increase their overall operational efficiency.

20. Kumar, V., & Reddy, S. (2019). “Analysis of Defence Procurement Reforms and Contractual Payment Mechanisms in India.” *International Journal of Social Science and Human Research*, Kumar and Reddy (2019) investigate changes in Indian defence purchasing with an emphasis on Hindustan Aeronautics Limited (HAL). The work describes the way HAL uses agreements-based financial methods like milestone payments, advance payments, performance-linked rewards, and penalties to guarantee the suppliers' adherence to the order and their on-time delivery. The study also indicates that well-organized payment conditions not merely facilitate the minimization of purchasing delays but also make the supplier's trust, teamwork, and operational friendliness, which is a step toward the practical implementation of the management of defence contracts.

3.2 STATEMENT OF THE PROBLEM

Hindustan Aeronautics Limited (HAL) is a top aerospace and defense institution in India, which takes care of the entire process from the design to the maintenance of aircraft, helicopters, and engines, along with their related systems. As a PSU under the Ministry of Defence, HAL is the keystone to the 'Force in readiness' that assures India's security, and also sets the stage for self-development in high-tech areas. Accomplishing these strategic goals through routine procurements and credit management is a must, buying and credit terms in defense contracts being the highlights of the areas concerned.

Procurement of raw materials, spare parts, and high-value components in large-scale defense manufacturing is mostly based on contracts with suppliers, who can be located both domestically and abroad. The feasibility of procurement processes hinges largely upon the existing purchase and credit policies, their degree of transparency, adaptability, and promptness. Moreover, a financial or contractual term should not be inefficient or vague, because otherwise, they might face a problem of production delays, which is further going to result in escalations of costs, and a break in the supply chain. In the case of HAL, the situation that it finds itself in due to these problems is a ripple effect of decrease in its operational efficiency and thus a negative impact over the long term. One of the regular problems that HAL comes across in its procurement system is the delay in payments to suppliers and contractors. In such situations, delays may occur because of procedural bottlenecks, complicated approval hierarchies, or strict observance of government financial regulations.

Furthermore, there is a possibility that ambiguous or obsolete purchasing policies might cause misunderstandings of contract obligations, delivery schedules, and penalty clauses. Defence contracts are in a situation where, if only a little, even slight inaccuracies, and unreliability of the underlying matter of the contract are the most critical issues leading the company to face a big operational disruption. The monetary sources of these issues are scattered far beyond HAL's internal area - they also impact the company's reputation, cooperation, and ability to meet the timelines critical for national defense.

It is very necessary to know the efficiency of the structure and implementation of purchase and credit policies of HAL after taking into account the enormous contribution made by HAL to the Indian defense ecosystem. Not only from the financial and managerial accounting perspective, but also from the strategic point of view, as the overall efficiency of the defense contract secretariat will be decided by it, these terms have to be evaluated. The identification of the current strengths and weaknesses of HAL's system can be instrumental in reforming the approach of fiscal discipline, increasing supplier satisfaction, and project delivery expediting.

This research is, therefore, the means for the defended contract purchase and credit terms in HAL. The research aims to ensure that it checks all the related problems that this nature such as delays, inflexibility of clauses, un... In addition, the study would also talk about the effect these troubles have on HAL's working capital, cash flow, and general organizational performance. The systematic purchase and credit theoretical evaluation is the study that by the facilitation of the indication of the spots where the reforms or changes may have pocketed most transparency, efficiency, and reliability of HAL's purchasing and payment processes.

In general, the purchase and credit features factoring in HAL's defence contracts is highly significant from both operational and financial perspectives. An adequately structured system promoting oneness, punctuality, and accountability in payment and procurement activities may be a real benefit for strategic qualities and the company's competitiveness further. Consequently, this study problem is first to resort back to the issue of dealing with the problem by thoroughly revising, dissecting and augmenting HAL purchase and credit apparatus to at least operations not being obstructed, healthier supplier relationships and better contract management efficiency.

From a financial management perspective, purchase and credit terms directly influence the liquidity position of HAL, the working capital cycle, and the budget utilization. A badly organized payment schedule can lead to the situation where inflows and outflows become unmatched thereby affecting the organization's ability to finance its ongoing operations without the need to borrow funds. This can, in the long run, decrease the level of profitability making financial reserves tighten and thus, in turn, impact the overall fiscal sustainability of HAL. Consequently, a thorough review of these terms is not simply a staff affair but a strategic necessity for HAL to secure long-term financial stability. Moreover, the performance of HAL purchase and credit terms is also a reflection of internal efficiency and the responsibility of the management apart from having mere financial

implications. The availability of open and adjustable credit policies makes it possible for decisions to be taken within the set timeframe, reliability of the supplier to be promoted and hence the better planning of production which in turn attracts the total utilization of the costs.

Such a study scenario will not only reveal the existing inefficiencies but also provide practical ways to rectify them. Hence, this work aims at closing the gap between formulating financial policies and putting them into practice, thereby helping HAL to carry out discipline of its spending without losing operational flexibility. The issue being researched through this study can be seen in how well HAL staggers its purchase and credit terms to help it secure defence contracts, manage supplier relationships, and maintain financial stability. Recognition and remedy of the problem areas that confront the organization such as problems in the payments, rigid terms, procedural inefficiencies and lack of flexibility will help create a procurement system that is more efficient, transparent, and reliable. Eventually, the information obtained from this evaluation will allow HAL to foster working capital management, instill supplier trust, and maximize overall performance in the highly competitive and strategically sensitive aerospace and defence sector.

3.3 NEED FOR THE STUDY

Hindustan Aeronautics Limited (HAL) is an organization that is most involved in aerospace and defence manufacturing in India and it is responsible for the whole process of the design, development, production, and maintenance of aircraft, helicopters, and related systems. A premier Public Sector Undertaking (PSU) governed under the Ministry of Defence, HAL is greatly instrumental in the country's self-reliance in defense manufacturing and also in the progress of technology. In order to project the on-time execution of the projects and retain the work standards that are being anticipated in defense manufacturing, the proper procurement systems and well-planned credit terms are of equal necessity.

The defense manufacturing industry is governed by government rules, financial regulations, and contractual obligations. HAL regularly gets into contracts with several vendors and suppliers to buy the materials that go into production. The purchase and credit terms here define what payments will be made, how credit will be allowed, and under what conditions will be checked for delivery and performance. Thus, the purchase and credit mechanisms have a very influential role to play in maintaining a flow of cash, handling working capital and, at the same time, vendor relations. This research is required due to the need for more financial discipline and operating efficiency in government enterprises, especially those active in the national sector, such as the defense sector. In the past, HAL has experienced difficulties over issues such as late payments, heavy approval procedures, and a lack of credit management. Such challenges, if left unaddressed, may put project schedules at risk of going haywire, leading to higher expenses and creating friction between HAL and its suppliers. In the context of a sector that operates on time and can be trusted, these delays can cause production backlogs that can affect the overall defense preparedness of the country.

This is also the case of a major reason for the study, namely the changes in defense procurement policies in India. The government of India has been implementing reforms to increase transparency, accountability, and efficiency in procurement and contract management. Hence, PSUs like HAL are required to maintain the parity of their internal policies and procedures with the alterations of these guidelines. Conducting an assessment of the current purchase and credit terms will also be instrumental in uncovering whether HAL's existing practices are in line with the modern contract management and financial governance standards. Furthermore, the direct and

indirect factors of HAL operations are decisions that must be made together with an analysis of how successfully the asset's purchase and credit terms have been put into action. The company operates on several contracts at the same time in different business areas, and each of them has its own payment method and contractual obligations. The lack of efficiency in one department can affect the financial management of the whole organization, and as a result, it will disrupt coordination among the other departments. Therefore, there is a demand for an evaluation of whether HAL's current systems are strong enough to manage such a level of complexity without losing efficiency or compliance.

Supplier relationship management is the perspective that this study is also very important. HAL's supply of the critically needed materials and components heavily depends on foreign as well as local vendors. A late payment or hard credit terms will incite the suppliers to be dissatisfied with an engagement in future transactions. Consequently, this will make it difficult for HAL to have access to good materials and technology cooperation. This study can be utilized for identifying the areas that need improvement in the company's payment and credit-related processes, which can reedify. The improvements can bring the company back to operational stability and this in turn will strengthen long-term vendor partnerships.

First of all, this study is also valuable for policymakers as well as management professionals who wish to streamline procurement systems at public sector undertakings. The scenario of HAL can be utilized as a model that aids other PSUs in similar sectors that are engaged in big-scale buyer activities and have intricate supply contract arrangements. These research results might help such institutions to employ the right payment frameworks that would lead to a decrease in operational bottlenecks and ensure the financial accountability of the organization.

With regards to globalization and the increasing competition in defense manufacturing, HAL has to assure that its internal financial systems are not only aligned with the regulatory requirements, but are also competitive and efficient. The assessment and upgrading of purchase and credit policies will result in the dedication of HAL to the following objectives: higher operational efficiency, better project coordination, and financial performance improved.

These findings of the study can influence other areas as well. For example, they can become a source of reference of other Public Sector Units and big companies that manufacture and take part in government bids. They would be able to create better procurement frameworks, conduct fiscal management with quality, and maintain the smooth operational execution of their organizations.

3.4 SCOPE OF THE STUDY

The scope of this study consists of the assessment and breakdown of the various buying and credit terms that are used in the contracts of Hindustan Aeronautics Limited (HAL) in the defence sector. Being the leader in aeronautics and the defence field in India, the contribution of HAL to the whole process of design, fabrication, and the after-service of aircraft, helicopters, engines, and the like, is very significant, plus it also plays a vital role in the associated systems.

Vendor and supplier organizations are the lifeline of the company which is dependent on them for meeting their operational and strategic needs in the Indian defense market. So, the financial management as well as the supply chain efficiency both require purchases and credit terms to be understood and made with the optimum level of terms.

This research is mainly centered around the contractual provisions in the purchasing and payment sections of HAL's procurement system in the defense sector. The goal here is to look into what purchase orders are being created, what payment contracts are usually agreed upon with vendors, and how these contracts in turn help the cash flow and liquidity of not only HAL but also its vendors. Besides the aforesaid aspects the study will also delve into budget releases linked to milestones, liquidated damages upon occurrence of delays, and rewards like discounts or priority allotments for on-time delivery and quality compliance that can be offered to sellers.

This work is limited to contracts dealing with defense equipment and services procured by HAL in the last few years. It will look at suppliers both resident in and outside the country, with the objective of providing help in understanding how credit terms can affect supplier performance, risk mitigation, and the efficiency of the procurement process as a whole. The research, however, will only briefly cover the basic principles of procurement management and will not go deeper into the detailed operational processes of the manufacturing or the technicalities of the products.

Currently, the study concentrates on the assessment of the buying and credit terms in defence contracts of Hindustan Aeronautics Limited (HAL) which is an outstanding Indian aerospace and defence company. HAL, set up under the Ministry of Defence, has a vital role in the research, production, and upkeep of the aircraft, helicopters, engines as well as other related defence systems. As the organization's buying process is critical to national security and technological self-

reliance, it is required to be not only effective but also transparent and well-defined. The defence contracts that are usually large-scale and involve very high-value transactions must ensure high efficiency in the purchase as well as credit terms to achieve financial discipline, avoid delays and make operations run smoothly. The primary objective of the study is to examine the purchasing terms that are adopted by HAL in their contracts, in detail. Besides this, it also aims to acquire information on various kinds of purchase orders which include standard purchase orders, long-term contracts and framework agreements along with company requirements under the particular clause. The investigation additionally explores the credit terms that are accessible for the vendors, for example, instalment schedules, credit periods, advances, progressive payments and the punishments or rewards that have been implemented. The study indicates these factors to emphasize the financial and organizational contributions to HAL as well as its partners. The efficient utilization of purchase and credit terms may even go as far as influencing the performance capacity of the suppliers, thereby, strengthening the trustworthiness of supply chains and lessening the probability of occurrence of delays or defaults.

This study is extensive in its examination of the Defense Buying Agency of HAL, tracing the purchasing of flying parts, power units, avionics, and maintenance services through various kinds of contracts. Understanding of credit terms across different procurement sources are provided by comparing domestic and international suppliers. Through the lens of procurement finance and contracts, however, the study refrains from technical specification or product manufacture process in the defense area.

Ultimately, the study aims to present a diverse range of reflective considerations including:

Face Analysis of Purchase Terms: The examination of the different types of purchase orders, contract clauses, supplier agreements, and obligations. On the other hand, the Evaluation of Credit Terms is the study of payment schedules, credit periods, milestone-based payments, penalties, and incentive structures. Influence of Financial Administration: Study of the main issues in case of purchase and credit terms in working capital, liquidity, and financial planning.

3.5 OBJECTIVES OF THE STUDY

1. To study the existing purchase and credit policies followed by HAL in its defence contracts.
2. To examine the impact of credit terms on HAL's financial performance and cash flow.
3. To identify any risks or delays caused by existing purchase and payment processes.

3.6 HYPOTHESES

This research is aimed at evaluating the purchasing and credit terms in Hindustan Aeronautics Limited (HAL) defence contracts and, accordingly, sets out the following assumptions:

Hypothesis 1: Purchase Terms Influence on Procurement Efficiency

Null Hypothesis (H_0): The terms of purchase in HAL's contracts have no significant impact on the efficiency of the procurement process.

Alternative Hypothesis (H_1): The terms of purchase in HAL's contracts have a significant impact on the efficiency of the procurement process.

Hypothesis 2: Impact of Payment Terms on HAL's Financial Performance and Cash Flow

Null Hypothesis (H_0): The financial performance and cash flow of HAL are not significantly affected by the payment terms and credit policies followed by HAL.

Alternative Hypothesis (H_1): The financial performance and cash flow of HAL have been significantly influenced by the payment terms and credit policies followed by HAL.

These hypotheses are verified by collecting data from HAL's procurement and supplier records, followed by analyzing the information to figure out the connection between credit/purchase terms and the total contract effectiveness.

3.7 RESEARCH DESIGN

1. Introduction

The research design is the base for any research work as it is basically a structured plan for conducting the research, and at the same time, it also assures the objectives of the study to be achieved in an efficient and effective manner. The “Evaluation of Purchase and Credit Terms in HAL’s Defence Contract” research design clarifies the procedure and the approach which have been chosen to understand the buying and credit policies prevailing at Hindustan Aeronautics Limited (HAL), a major Public Sector Undertaking under the Ministry of Defence, Government of India. This research design is the whole work’s master plan. It specifies the ways in which information will be collected, analyzed, and interpreted for assessing the efficiency of the purchase and credit terms of HAL. As the objective of the study is to present the existing practices and their efficiency evaluation without any variables being changed, the use of a descriptive research design has been found to be the best method.

2. Type of Research Design

The research implementation in this case study is the descriptive research design, which is mainly concerned with providing the most accurate description of the current situation, without the establishment of cause-effect relationships. Descriptive research can be used to get a grasp of the existing conditions, the processes, and the relationships in a more formalized way.

Such a design allows the researcher to:

- Find out what purchase and credit terms, among others, have been implemented at HAL.
- Determine to what extent these terms have provided support to the company’s financial and contractual performance.
- Discover if there are any issues or gaps with the current system.

- Offer suggestions that may lead to the increase of the overall productivity of the procurement and payment process.

So, the descriptive design is appropriate for this study as it depends on the factual and analytical assessment of HAL's defence contract procedures.

3. Objectives of the Research Design

The leading purposes of this research methodology are as follows:

- Detailing and comprehending how HAL utilizes purchasing and credit policies for defense contracts.
- Estimating the effectiveness and sufficiency of these agreements' terms.
- Uncovering possible obstacles or inefficiencies in the current purchasing and payment system.
- Suggesting ways to increase HAL's management of contracts and credit through the provision of advice.

Such goals keep the research grounded, methodical, and pertinent to the institution of HAL.

4. Nature of the Study

The study combines aspects of both qualitative and quantitative nature.

- Qualitative Aspect: To comprehend HAL's procurement framework, contract clauses, and payment procedures, the study gathers information through reviewing documents and consulting experts.

- **Quantitative Aspect:** It involves processing of numbers relating to payment timelines, credit durations, and contract value patterns to measure the level of efficiency and effectiveness, if the data is available.

The use of this combination enables the researchers to get a complete understanding of the problem they are researching.

5. Scope of the Research Design

The scope of this study is limited to the assessment of the buying and credit terms of HAL defense contracts. The emphasis is on gaining insight into the impact of these terms on payment efficiency, supplier relationships, and general operational performance. Besides these, the research does not extend to any other department like pricing strategy, cost estimation, or production processes.

Conclusion

To conclude, the developed research design in this article lays out a very systematic and well-laid plan for evaluating the buying and credit terms of HAL. The study, basing on a descriptive framework, allows a balanced treatment of both the realities of the organization and the strategic, by using mixed methods - qualitative and quantitative. Through the design, the authors are now able to provide the basis for making research findings on the aspects of the HAL system of defense contract management being convenient and efficient. To sum up, the research design created for the “Evaluation of Purchase and Credit Terms in HAL’s Defence Contract” study acts as an organized and rational guide that leads the way to accomplishing the research objectives efficiently. Employing a descriptive research design, the investigation clarifies the existing practices of purchasing and credit policies at Hindustan Aeronautics Limited. Moreover, the study uses a qualitative and quantitative approach, which gives a complete, unbiased, and fact-based account of the situation.

Various authentic and reliable sources like annual reports, manuals, and government publications have been used for secondary data collection, which have been carefully analyzed to check the efficiency and trustworthiness of HAL’s credit and procurement systems. Such a layout not only facilitates the recognition of the organization's credit framework's current strengths and the areas where it lacks but also provides a basis for the informed recommendations that can lead to operational and financial management efficiency.

In essence, the research design is a pivotal framework that bridges the gap between theory and practical evaluation, thus it ensures the study is focused, up-to-date, and beneficial from both academic and organizational viewpoints.

3.8 TOOLS FOR DATA COLLECTION

Secondary Data Collection Tools

The study on the “Evaluation of Purchase and Credit Terms in HAL’s Defence Contract” is a case study that utilizes secondary data. Secondary data is the information that has been previously gathered, organized, and published by various institutions. In order to understand, recognize, and evaluate the existing practices, etc., this dataset is used without any direct contact with the respondents.

In a major way, secondary data contributes to the success of this research study as it conveys true, and traceable information regarding HAL’s purchasing policies, contractual terms, and financial management systems. Owing to the nature of secrecy surrounding defense contracts, it is out of the question to get new original data directly from the officials of the internal department. Consequently, the research depends on accurate, open-access documents and published studies that describe procurement and payment mechanisms in the defense sector.

The main sources of secondary data used in this study include:

1. H.A.L. Annual Reports and Financial Statements: H.A.L. Annual Reports and Financial Statements provide in-depth data on the company’s operational performance, procurement activities, contract obligations, and payment practices. One can use these documents to gauge the financial stability and the purchase and credit system design of the organization.
2. Research Papers and Academic Journals: One can get theoretical and analytical insights for procurement efficiency, contract management, and credit policy evaluation by reading scholarly articles and research papers that are available through Google Scholar, ResearchGate, and other academic databases. Besides, these studies also furnish comparative perspectives between various public sector organizations.
3. Official Websites and Public Disclosures: The data related to tenders, procurement procedures, and supplier payment guidelines has been collected from official websites of HAL (www.hal-india.co.in and Ministry of Defence portals. These internet-based sources offer the latest and validated information that back the study.

The data from various sources were scrutinized, cross-verified, and interpreted to maintain their accuracy and consistency. To extract the relevant information and to identify the key patterns in purchase and credit terms, content analysis, and comparative review methods were employed. This research, as a whole, would be less reliable attainable if solely relying on primary data. To sum up, the use of secondary sources has given a strong base for the assessment of the purchase and credit terms in HAL's defence contracts. The findings that are based on the use of verified and published sources are factual, objective, and even close to the real-world practices of HAL and the Indian defence procurement framework.

3.9 LIMITATIONS OF THE STUDY

Every research can, despite all the efforts to be detailed and effective, have small or sometimes big limitations. The essence of referring to these limitations is an important part of the research process because it allows the readers to comprehend the extent of the study results within the given limitations. The work entitled "Evaluation of Purchase and Credit Terms in HAL's Defence Contract" is, indeed, subject to such acknowledgments. The mentioned problems arise as the study seeks to provide detailed analyses on HAL's purchase and credit policies and as it is limited by methodology, data sources, and organizational constraints. The foremost limitation of the work is the study, which is totally based on secondary data. It refers to the data that has already been collected, compiled, and published by researchers, organizations, or governments. The researcher in this study depended on HAL's annual reports, government manuals, policy documents, audit reports, and research articles.

Secondary data can open new vistas of knowledge and be a sturdy base for the analysis; still, it is accompanied by some limitations. The data that were gathered for some other reasons may not provide enough details on the research objectives. Also, secondary data is sometimes put together for certain periods, so some pieces of information may not be the most recent ones. This constraint bars the research from reflecting the latest changes in HAL's procurement and credit practices.

One more significant limitation is the degree to which access to confidential information is restricted. HAL, as a defence public sector undertaking, MAC highly confidential in its procurement, contract, and credit functions. Numerous details regarding individual contracts, payment schedules, and supplier agreements are at such a high level of classification that they are

not open to the public. As a result, the study would not be able to obtain the first-hand information of employees, managers, or contract officials. This limitation results in the study being bound by a certain type of data, i.e., those which are publicly available, and this may not demonstrate fully the internal challenges, procedural bottlenecks, or practical nuances in the implementation of purchase and credit terms at HAL. Time limitations are another factor that influences the study and are usually present in academic research as a common limitation.

Within the framework of the semester timeline for the study, the researcher had a limited amount of time to accomplish the task of thoroughly reviewing all the historical contracts and also cross-checking the various data sources. A comparative study with other public sector or private organizations that would have been quite extensive was also limited due to the available time. Thus, the study's findings and analysis are based on the most recent and accessible data and, therefore, may only partly reflect the long-term trends or variations in HAL's purchase and credit practices.

In the end, the research is limited to purchasing and credit terms in HAL's defense contracts only. Besides these, areas such as supplier relationship management, pricing strategies, contract negotiation techniques, and technical specifications of the procured items were not covered in the research.

This limited setting, on the one hand, helps the researchers to stay focused on their objectives and, on the other hand, restrains the extent of the study. Moreover, the way secondary data are interpreted is depending on the researcher which could lead to biases when dealing with complex procurement policies or contractual clauses.

As a result, although this research contributes a thorough breakdown of the purchase and credit terms of HAL, it is still essential to bear in mind the limitations incorporated by the dependence on secondary data, restricted access to confidential information, short time span, and the focus of the study. Knowing these limitations permits the readers to draw conclusions cautiously. Conceding these limitations, the study still affords the worth of the insights into the efficiency, organization, and effectiveness of HAL's purchase and credit systems and sets a precedent for subsequent investigations that may consider primary data collection or wider comparative analysis with other entities.

CHAPTER – 04

DATA ANALYSIS AND INTERPRETATION

4.1 DETAILS OF TOOLS USED FOR DATA ANALYSIS

Research data analysis forms a major component of any inquiry, as it facilitates the transformation of unprocessed details into valuable understandings. One way this can be seen is from the present study "Evaluation of Purchase and Credit Terms in HAL's Defence Contract" which has relied on secondary data sourced from HAL's annual reports, audit reports, government manuals, and published articles. This data has been subjected to analysis using a number of instruments such as Microsoft Excel.

➤ Microsoft Excel

Microsoft Excel is one of the most powerful spreadsheet tools that operate. It In a researchers can use it to organize, manipulate, and visualize data their data efficiently. In this research, the tool has been used with the goal of entering data, cleaning, and performing a preliminary analysis including calculation of sums, averages, percentages, and simple descriptive statistics. Besides, Excel is also capable of creating different charts and graphs such as clustered column charts, line charts, and pie charts which facilitate the visual representation of the trends in purchase and credit terms. The pivot tables attribute of this program allows quick summarizing of extensive datasets which in turn facilitates drawing of substantial insights from HAL's financial and operational records.

➤ SPSS (Statistical Package for the Social Sciences)

SPSS is an advanced statistical software that enables deep data analysis. In combining with the use of SPSS, the present study carried descriptive statistics, correlation, and regression analysis to comprehend the interactions of purchase values, credit terms, delivery periods, and payment delays in HAL's defence contracts. Moreover, the outputs generated by SPSS, such as the tables and significance values, are instrumental in confirming the hypotheses and facilitating data-driven decision-making.

4.2 DATA ANALYSIS AND INTERPRETATION

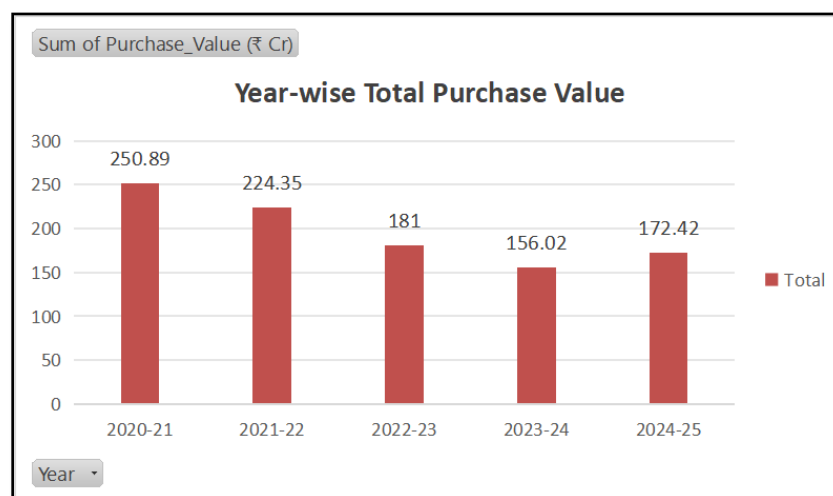
- **DESCRIPTIVE ANALYSIS**

1. Demonstrating Year-Wise Procurement Value of HAL (2020-2025)

These data are a reflection of HAL's procurement trends and fluctuations in their yearly expenditures, thus they give an understanding of the organization's purchasing patterns, budgeting strategies, and operational changes over time, besides showing the parts where the costs have been optimized and the resources allocated.

Table 4.2.1 Demonstrating Annual Purchase Value of HAL (2020-25)

Year	Sum of Purchase Value (₹ Cr)
2020-2021	250.89
2021-2022	224.35
2022-2023	181
2023-2024	156.02
2024-2025	172.42
Total	984.68



Graph 4.2.1 Demonstrating Annual Purchase Value of HAL (2020-25)

ANALYSIS AND INTERPRETATION

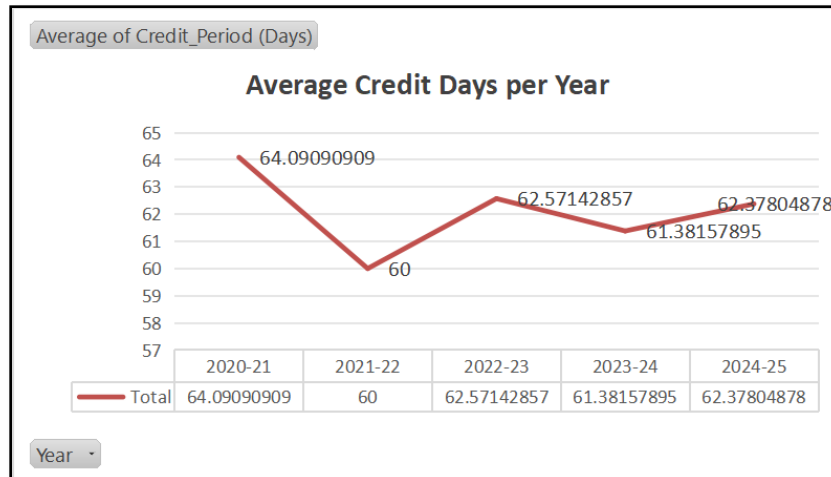
The bar chart displays HAL's total purchase value annually from 2020-21 to 2024-25. The greatest purchase value was in 2020-21 (₹250.89 Cr) which was then followed by a steady decrease to ₹156.02 Cr in 2023-24 and a marginal increase to ₹172.42 Cr in 2024-25. The mean purchase value during the five years is ₹196.94 Cr, the median that represents the middle value in the ordered dataset is ₹181 Cr. The standard deviation (SD), which reflects the spread of the purchase values, is ₹39.34 Cr. By looking at the graph, one can understand that HAL's buying of goods and services has gone down gradually from 2020-21 to 2023-24 which is a sign of operational adjustments or cost reduction activities. A small increase in 2024-25 indicates a slight return to the procurement. The mean being above the median suggests that the first years had higher purchase values which resulted in the average going up. The standard deviation is indicative of moderate variability which means that the procurement activity was generally consistent but with slight fluctuations during the period.

2. Demonstrating Year-Wise Average Supplier Credit Periods

The average supplier credit periods were quite similar for the different years, thus showing that the payment terms and the trust of the suppliers were stable. Small fluctuations imply that changes were made according to the requirements of the project or the results of the negotiations. In general, the stability is a manifestation of proper financial planning, payment of dues on time, and the organization's ability to maintain good relationships with suppliers.

Table 4.2.2 Demonstrating Average Supplier Credit of HAL (2020-25)

Year	Average of Credit Period (Days)
2020-2021	64.09090909
2021-2022	60
2022-2023	62.57142857
2023-2024	61.38157895
2024-2025	62.37804878
Total	62.1



Graph 4.2.2 Demonstrating Average Supplier Credit of HAL (2020-25)

ANALYSIS AND INTERPRETATION

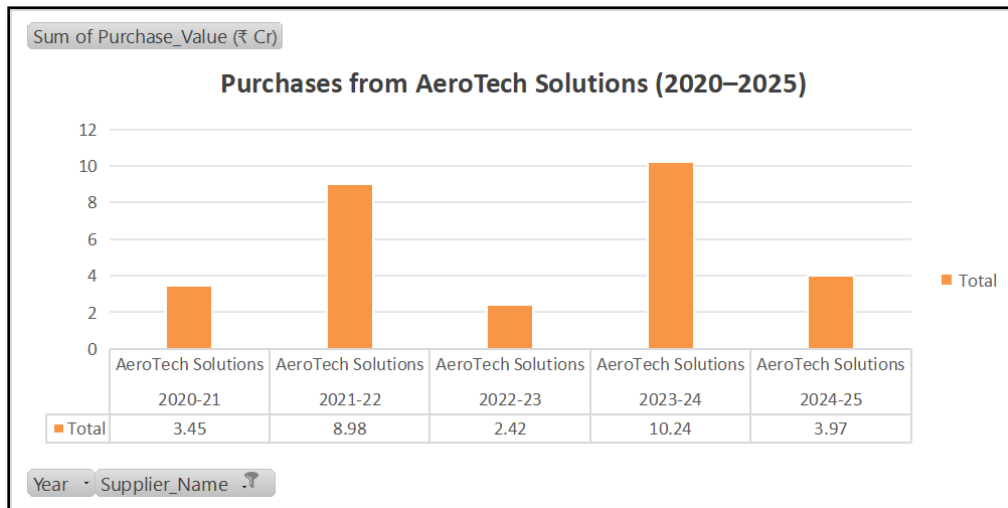
The analysis of the average supplier credit period over a five-year horizon from 2020–21 to 2024–25 reflects a balanced scenario with marginal fluctuations between years. Both the average and median credit periods hover around 62 days (62.10 days for the mean and 62.38 days for the median), pointing to a distribution of credit days that is nearly symmetrical. The low standard deviation of 1.36 days indicates that supplier payment terms have been very similar during the five years and hence, there is minimal variation in them. The longest average credit period was in 2020–21 (64.09 days), and the shortest one was in 2021–22 (60 days). The small range between these two values shows that suppliers have been issuing steady and predictable credit terms. The stability in such credit terms indicates supplier relationships in good health, payment practices on time, and financial planning by the organization done efficiently. The differences to a small extent in the years may be due to projects and contracts negotiated changes. In summary, the steady credit period trend is a good indicator of the organization's financial reliability and operational discipline.

3. Demonstrating Year-Wise Purchases from AeroTech Solutions (2020-2025)

The information here illustrates the annual transactions with AeroTech Solutions over the years 2020-2025. It emphasizes procurement changes, pointing to times of increased expenditures and times when fewer purchases were made. Studying these trends is useful for comprehending buying behavior, figuring out how much one is reliant on a supplier, and preparing for budgeting and stock control purposes.

Table 4.2.3 Purchase Value Analysis of AeroTech Solutions (2020–2025)

Year	Supplier Name	Sum of Purchase Value (₹ Cr)
2020-2021	AeroTech Solutions	3.45
2021-2022	AeroTech Solutions	8.98
2022-2023	AeroTech Solutions	2.42
2023-2024	AeroTech Solutions	10.24
2024-2025	AeroTech Solutions	3.97
Total		29.06



Graph 4.2.3 Purchase Value Analysis of AeroTech Solutions (2020–2025)

ANALYSIS AND INTERPRETATION

The stacked column chart showing purchases from AeroTech Solutions over 2020–21 to 2024–25 illustrates a change in the buying pattern. The first year 2020–21 saw purchases of ₹ 3.45 Cr followed by a peak of ₹ 10.24 Cr in 2023–24 and a decrease to ₹ 3.97 Cr in 2024–25. The average purchase amount of ₹ 5.81 Cr is representative of the yearly purchases during the period while the median value of ₹ 3.97 Cr is indicative that in most years purchases were below the average. The range of purchases from AeroTech Solutions to HAL as uncovered by a standard deviation of ₹ 3.17 Cr demonstrates that they have been highly dissimilar and therefore not in a trend of

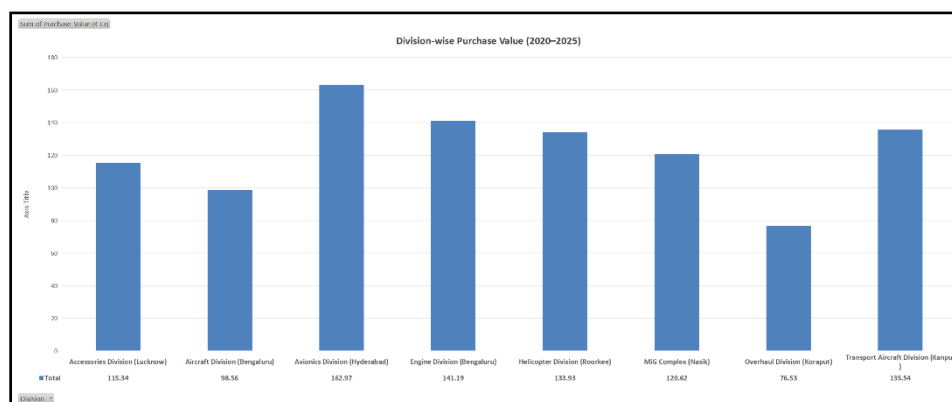
continuity. Such a fluctuation may be caused by irregular project requirements, contract-based procurement, or supply cycle variations.

From a business point of view, the evidence indicates that HAL's collaboration with AeroTech Solutions depends on the projects, purchases surge during periods of high demand. To achieve efficiency, HAL can resort to long-term supply agreements or forecast-based planning to stabilize purchase volumes. In essence, the chart is a story of the five years that tell of tumultuous procurement trends, but with the influence of strategy, they become less unpredictable.

4. Demonstrating Division-wise Purchase Value of HAL (2020-2025)

Table 4.2.4 Demonstrating Division-wise Purchase Value of HAL

Division	Sum of Purchase Value (₹ Cr)
Accessories Division (Lucknow)	115.34
Aircraft Division (Bengaluru)	98.56
Avionics Division (Hyderabad)	162.97
Engine Division (Bengaluru)	141.19
Helicopter Division (Roorkee)	133.93
MiG Complex (Nasik)	120.62
Overhaul Division (Koraput)	76.53
Transport Aircraft Division (Kanpur)	135.54
Grand Total	984.68



Graph 4.2.4 Demonstrating Division-wise Purchase Value of HAL

ANALYSIS AND INTERPRETATION

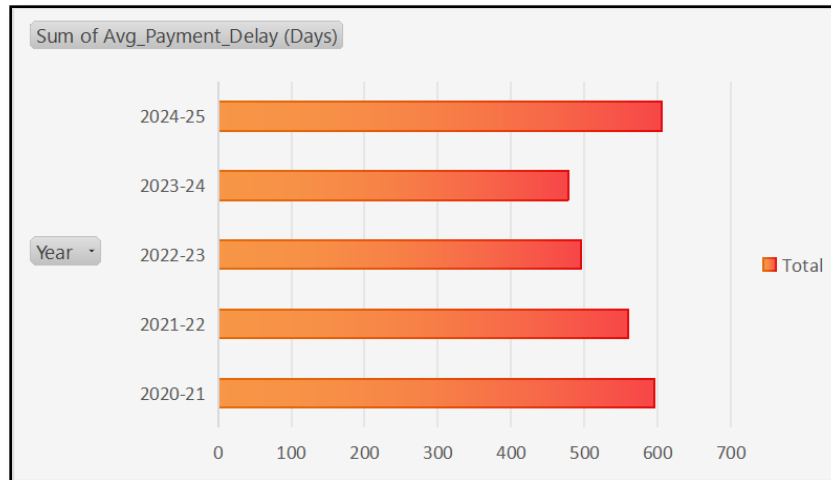
The clustered column chart represents the purchase values of HAL across the eight divisions from 2020–25. The Avionics Division (Hyderabad) was the one with the maximum purchase value (₹162.97 Cr), whereas the Overhaul Division (Koraput) was the lowest (₹76.53 Cr). The average purchase value across divisions is ₹123.09 Cr, and the median is ₹127.28 Cr, which suggests that the majority of divisions' purchases are close to this range. The chart with a standard deviation of ₹25.04 Cr depicts the changes to a moderate extent among the divisions — a few divisions such as Avionics and Transport Aircraft have considerably higher purchase volumes, whereas smaller divisions like Overhaul have lower values. In general, the chart reveals division-specific procurement patterns, which facilitates the understanding of budget allocation, supplier management, and strategic planning for HAL's operations.

5. Demonstrating Sum of Avg Payment Delay (Days)

The changes in the average time of payment are captured in the data, reflecting how fast the payment process was carried out. In periods of higher delays, payments were generally slow and thus, those periods could have been affected by blockages in the operations or difficulties in the administrative side of the process. On the other hand, lower delays signal that management has become more efficient and payments to suppliers have been made on time.

Table 4.2.5 Demonstrating Sum of Avg Payment Delay (Days) (2020-2025)

Year	Sum of Avg Payment Delay (Days)
2020-2021	595
2021-2022	560
2022-2023	496
2023-2024	478
2024-2025	606
Total	2735



Graph 4.2.5 Demonstrating Sum of Avg Payment Delay (Days) (2020-2025)

ANALYSIS AND INTERPRETATION

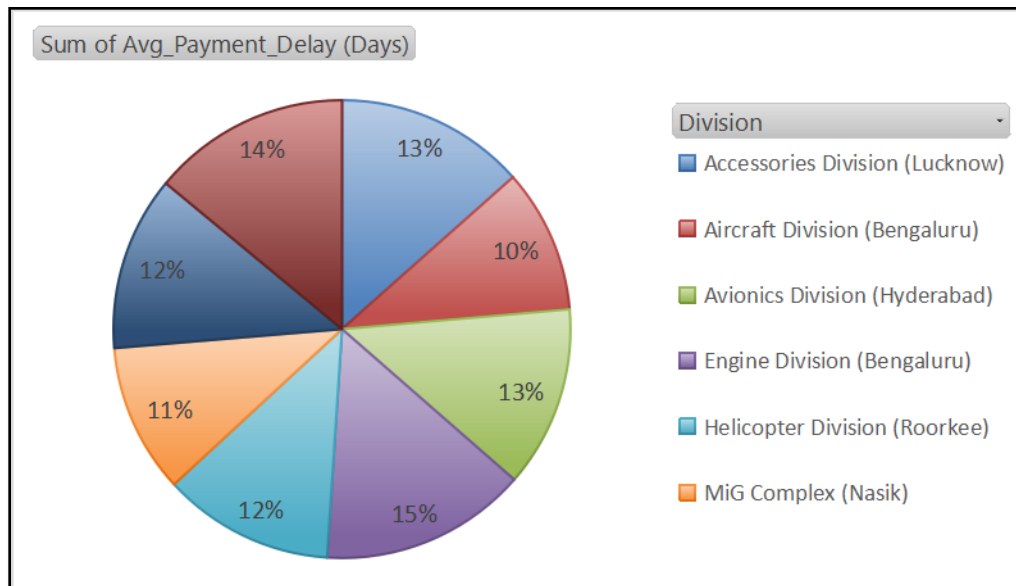
The clustered bar chart explains the total average payment delay days from 2020–21 to 2024–25. The average payment delay over these five years is 547 days, the median being 560 days, and the standard deviation is around 58 days, which shows a moderate range of variability in delays. From the chart, it is evident that 2024–25 was the year with the longest total payment delay (606 days), thus, slower payment processing, while 2023–24 was the year with the shortest total payment delay (478 days), therefore, faster payment processing. The changes between the years show that there were irregularities in payment cycles due to which could be delayed operations, supplier-specific factors, or internal processing efficiency. Keeping an eye on these trends is very helpful in locating the times of bottlenecks and in making plans for cash flow management.

6. Demonstrating Sum of Avg Payment Delay by Division (2020–25)

The section is all about the average total number of days per year each division of HAL took to process payments over 2020-2025. The figures are used to find out those divisions that have the longest payment delays of their cash flow and to compare their general payment performance within the organization.

Table 4.2.6 Demonstrating Sum of Average Payment Delay by Division

Division	Sum of Avg Payment Delay (Days)
Accessories Division (Lucknow)	368
Aircraft Division (Bengaluru)	278
Avionics Division (Hyderabad)	350
Engine Division (Bengaluru)	400
Helicopter Division (Roorkee)	331
MiG Complex (Nasik)	288
Overhaul Division (Koraput)	337
Transport Aircraft Division (Kanpur)	383
Grand Total	2735



Graph 4.2.6 Demonstrating Sum of Average Payment Delay by Division

ANALYSIS AND INTERPRETATION

The pie chart gives an overview of the total average payment delays by each division over the years 2020-2025. It has been shown that the average payment delay for all divisions is around 341.88 days, with the median delay being 343.5 days, which implies that most of the divisions have their delays around this central value. The standard deviation of 41.54 days is an indication

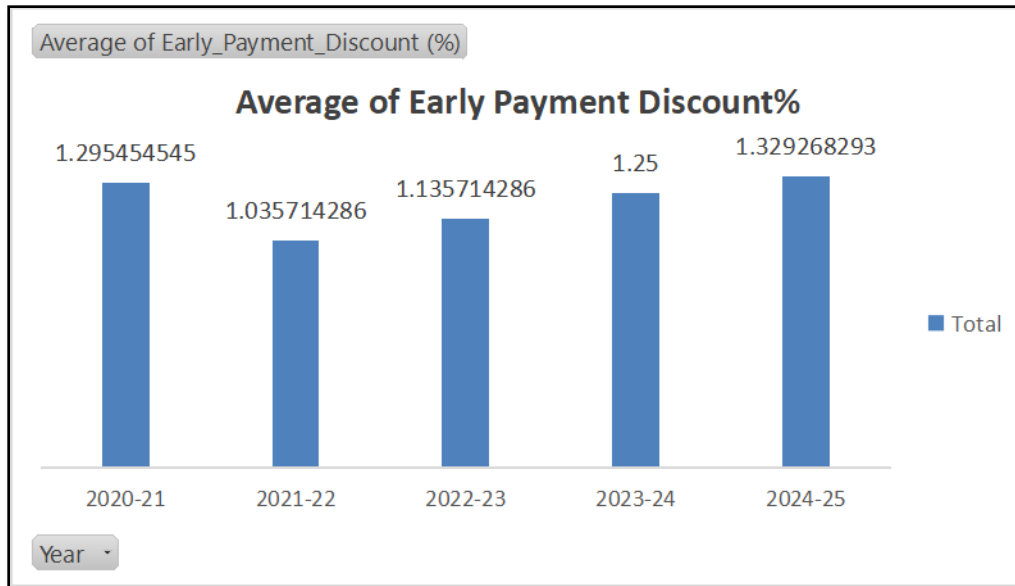
of a moderate range of payment delays among divisions. The silhouette of the engine department (Bengaluru) can be seen in the pictorial representation of the data which suggests that it records the highest contribution of the delayed payment to the total with a delay of 400 days, a long way Transport Aircraft Division (Kanpur) is behind with the second-highest delay, whereas the Accessories Division (Lucknow) is in third place. Until recently the Aircraft Division (Bengaluru) was the one with the least delay of 278 days which is an indication of payment efficiency that is relatively high. In essence, the graphic points to the divergence of the payment execution between divisions and such differences can committedly be the driving force that results in variegated payment efficiency being stacked in HAL.

7. Demonstrating Mean Early Payment Discount by Year (2020–2025)

The average percentage of early payment discounts enjoyed by HAL over a period from 2020 to 2025 is explained in this section. It throws light on the payment efficiency of HAL to its suppliers and the extent to which the organization has been able to take advantage of the early payment discount benefits throughout the period.

Table 4.2.7 Demonstrating Mean Early Payment Discount by Year (2020-2025)

Year	Average of Early Payment Discount (%)
2020-21	1.295454545
2021-22	1.035714286
2022-23	1.135714286
2023-24	1.25
2024-25	1.329268293
Grand Total	1.21125



Graph 4.2.7 Demonstrating Mean Early Payment Discount by Year (2020-2025)

ANALYSIS AND INTERPRETATION

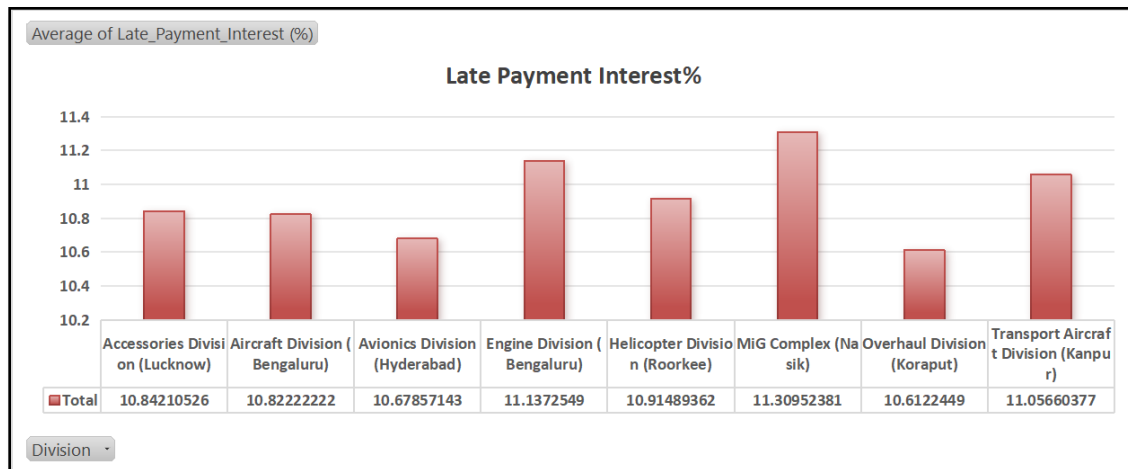
The clustered column chart illustrates the Mean Early Payment Discount (%) by Year for the years 2020–2025. It indicates that HAL's average early payment discounts have been quite stable over the years with just minor changes. The average discount rate over all years is 1.21%, the median is 1.25%, and the standard deviation of 0.11% shows that there is very little change. First of all, the 2024–25 year recorded the maximum discount (1.33%), thus it can be inferred that the payment of the obligations was done quite timely during that period, while 2021–22 had the lowest (1.04%), hence it could be a small amount of delay in that period. In general, the chart is emphasizing the consistent performance of HAL in obtaining early payment discounts, which is a good indication of financial discipline having been maintained throughout the years.

8. Demonstrating Average Late Payment Interest by Division

The study displays the average late payment interest that has varied across different divisions. The Engine Division, on the other hand, is the one that slightly elevates the interest rate for delayed payments, whereas the Avionics Division is the one that, on average, has the lowest interest rate.

Table 4.2.8 Average Late Payment Interest by Division (2020-2025)

Division	Average of Late Payment Interest (%)
Accessories Division (Lucknow)	10.8421
Aircraft Division (Bengaluru)	10.8222
Avionics Division (Hyderabad)	10.6785
Engine Division (Bengaluru)	11.1372
Helicopter Division (Roorkee)	10.9148
MiG Complex (Nasik)	11.3095
Overhaul Division (Koraput)	10.6122
Transport Aircraft Division (Kanpur)	11.0566
Grand Total	10.9125



Graph 4.2.8 Average Late Payment Interest by Division (2020-2025)

ANALYSIS AND INTERPRETATION

The clustered column chart representing fields of different divisions in case of late payment interest is a great tool to visually compare the different interest rates that the divisions have applied. On average, the interest rate is close to 10.92%, which tells that in most divisions the same interest policy has been applied. The median of 10.88% that is very close to the mean also indicates that the distribution is quite balanced and there are no significant deviations or extreme values. The standard deviation, which is only 0.22%, confirms even more the closeness of the interest rates to

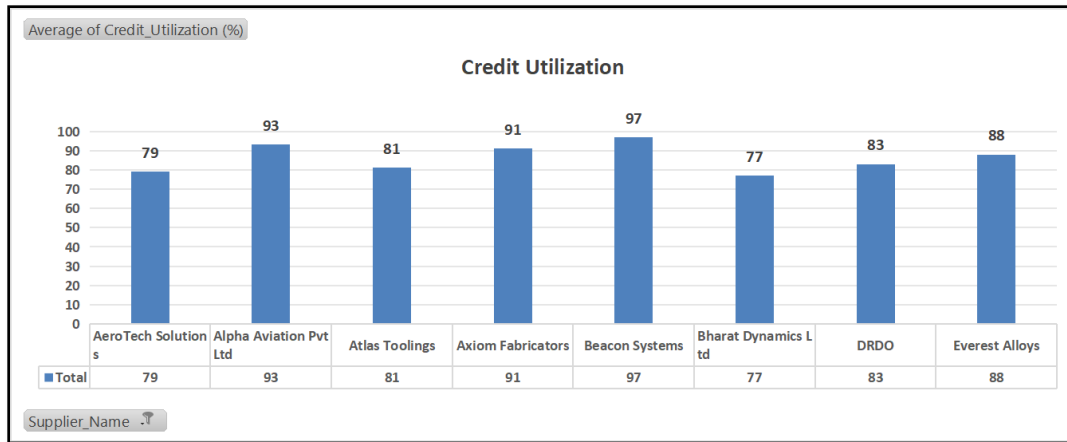
each other in the different divisions since the clustered columns are nearly the same height meaning they have similar values. The MiG Complex, with an interest rate of 11.31%, is the highest, and the Overhaul Division, with an interest rate of 10.61%, is the lowest, but the overall difference is very small and, thus, late payment interest is quite regular and predictable. The present visualization is very efficient in showing how the divisions differ in terms of relative figures while at the same time, the general uniformity of the rates is most evident.

9. Demonstrating Supplier-wise Average Credit Utilization (%)

The analysis presented here reflects the effectiveness of each supplier in the use of credit facilities extended to them by the company. It is an instrument for measuring the suppliers' financial discipline and their regularity in making payments. A higher utilization ratio reflects a more efficient use of the available credit limits, whereas a lower utilization ratio may imply that the users are adopting a more cautious or that they are delaying their usage. This information is thus instrumental in the supplier performance evaluation processes and financial planning becoming more effective.

Table 4.2.9 Supplier-wise Average Credit Utilization (%)

Supplier Name	Average of Credit Utilization (%)
AeroTech Solutions	79
Alpha Aviation Pvt Ltd	93
Atlas Toolings	81
Axiom Fabricators	91
Beacon Systems	97
Bharat Dynamics Ltd	77
DRDO	83
Everest Alloys	88
Grand Total	85.03



Graph 4.2.9 Supplier-wise Average Credit Utilization (%)

ANALYSIS AND INTERPRETATION

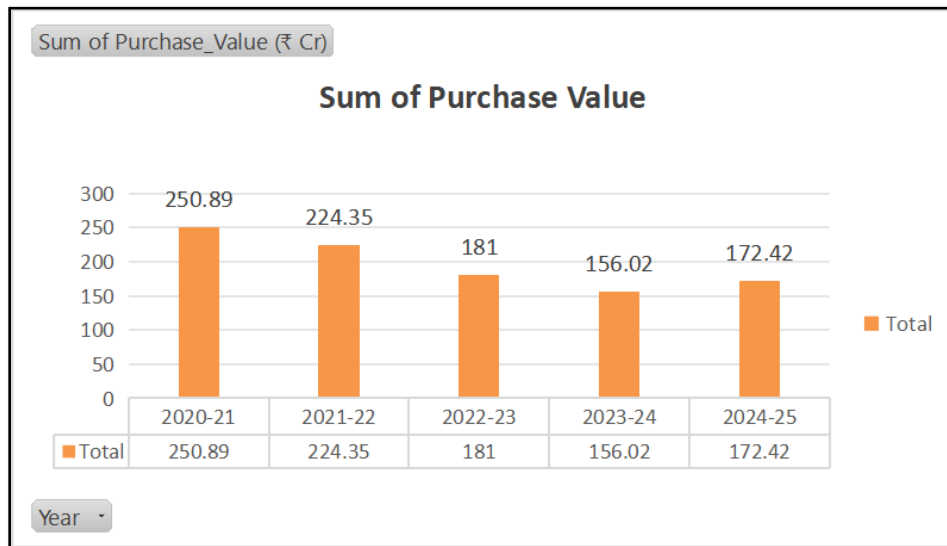
The clustered column chart that depicts supplier-wise average credit utilization (%) helps in understanding how each supplier has used the credit facilities provided by the company. As per the analysis, the majority of suppliers have shown to be good users of credit with an overall mean of 86.13%. Beacon Systems is the one that utilizes the credit facilities provided by the company in the most excellent way with a figure of 97%, thereby reflecting very good financial stability, whereas Bharat Dynamics Ltd is at 77%, which is showing that a more cautious approach to credit has been taken. The standard deviation of 7.18 signifies that there is a moderate range of variation among suppliers, which means that although utilization levels change slightly, most suppliers are in a position to maintain strong and consistent credit performance. In general, the clustered column chart has been successful in displaying the suppliers' performance in managing credit resources.

10. Demonstrating Overall Year-Wise Procurement Growth Trend

The Annual Procurement Growth Trend analysis reveals the changes in procurement values over the years very vividly. After comparing total purchase values from 2020–21 to 2024–25, it is possible to recognize the development or decrease patterns. Such a kind of report enlightens the company one of the spending behavior, supplier engagement, and procurement efficiency over time.

Table 4.2.10 Year-Wise Procurement Growth Trend

Year	Sum of Purchase Value
2020-21	250.89
2021-2022	224.35
2022-2023	181
2023-2024	156.02
2024-2025	172.42
Total	984.68



Graph 4.2.10 Year-Wise Procurement Growth Trend

ANALYSIS AND INTERPRETATION

The procurement trend, year-wise, from 2020-21 to 2024-25, presented through a clustered column chart, demonstrates a decline very distinctly from 250.89 Cr in 2020-21 to 156.02 Cr in 2023-24, and then a slight comeback to 172.42 Cr in 2024-25. The average procurement of 196.94 Cr is indicative of the procurement value over the five years on average, whereas the median of 181 Cr is the middle point of the dataset, thereby implying that in half of the years the procurement was

above this value and in the other half below. The standard deviation of 35.19 Cr is a measure of the fluctuation at a moderate level, it indicates that there is a substantial difference between the highest and the lowest procurement years. The clustered column chart serves as a handy tool to compare each year's procurement visually and it is very evident from the downward trend as well as from the slight rebound in the last year.

- **Objective 2: To examine the impact of credit terms on HAL's financial performance and cash flow.**

Null Hypothesis (H_0): The financial performance and cash flow of HAL are not significantly affected by the payment terms and credit policies followed by HAL.

Alternative Hypothesis (H_1): The financial performance and cash flow of HAL have been significantly influenced by the payment terms and credit policies followed by HAL.

Correlations						
		Payment_Terms (Days)	Credit_Period (Days)	Delay (Days)	Avg_Payment_Delay (Days)	Credit_Utilization
Payment_Terms (Days)	Pearson Correlation	1	-.012	.049	-.037	.032
	Sig. (2-tailed)		.809	.327	.459	.525
	N	400	400	400	400	400
Credit_Period (Days)	Pearson Correlation	-.012	1	-.037	.086	-.093
	Sig. (2-tailed)	.809		.461	.085	.064
	N	400	400	400	400	400
Delay (Days)	Pearson Correlation	.049	-.037	1	-.001	-.029
	Sig. (2-tailed)	.327	.461		.990	.562
	N	400	400	400	400	400
Avg_Payment_Delay (Days)	Pearson Correlation	-.037	.086	-.001	1	-.052
	Sig. (2-tailed)	.459	.085	.990		.300
	N	400	400	400	400	400
Credit_Utilization	Pearson Correlation	.032	-.093	-.029	-.052	1
	Sig. (2-tailed)	.525	.064	.562	.300	
	N	400	400	400	400	400

(Table 4.2.11 H_1 Correlations)

INTERPRETATION

The correlation matrix represented the strength of association between different financial variables - Payment Terms, Credit Period, Delay, Average Payment Delay, and Credit Utilization. The

results of the analysis depict that most of these relationships are very weak and statistically insignificant. To illustrate, Payment Terms and Credit Utilization are weakly positively correlated (0.032), indicating that the credit utilization hardly changes with payment terms. In a similar way, Credit Period and Credit Utilization reveal a weak negative correlation (-0.093), however, the relationship is not significant.

The results, in fact, point to the absence of any strong links between these variables. Each factor, i.e., payment terms, credit period, or delays, was found to be quite independent of each other with almost no interinfluence.

Coefficients^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	84.707	1.986		42.661	<.001
	Payment_Terms (Days)	.015	.025	.031	.616	.538
	Credit_Period (Days)	-.036	.020	-.090	-1.792	.074
	Delay (Days)	-.010	.015	-.034	-.678	.498
	Avg_Payment_Delay (Days)	-.094	.109	-.043	-.859	.391

a. Dependent Variable: Credit_Utilization

(Table 4.2.12 H1 Coefficients)

INTERPRETATION

The regression analysis shows us the influence of different factors such as Payment Terms, Credit Period, Delay, and Average Payment Delay on Credit Utilization. According to the table, it appears that these variables do not have any significant impact on Credit Utilization as all the significance (Sig.) values are above 0.05. As for the variables, Credit Period (Days) has a slight negative effect on Credit Utilization (B = -0.036, Sig. = 0.074), which means that an extension of the credit period will lead to a slight decrease in credit utilization — however, this relationship is not statistically significant. Likewise, Payment Terms, Delay, and Average Payment Delay also indicate weak effects, which means that changes in these factors do not have a strong impact on the level of credit utilization. Put simply, the results are that Credit Utilization is hardly ever changed by payment or

credit terms and delays. This means that there could be other factors that are not accounted for in this model which have a greater influence on Credit Utilization.

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	404.430	4	101.107	1.257	.286 ^b
	Residual	31767.148	395	80.423		
	Total	32171.577	399			

a. Dependent Variable: Credit_Utilization

b. Predictors: (Constant), Avg_Payment_Delay (Days), Delay (Days), Payment_Terms (Days), Credit_Period (Days)

(Table 4.2.13 H1 ANOVA)

INTERPRETATION

The ANOVA table is instrumental in determining whether the four independent variables — Average Payment Delay (Days), Delay (Days), Payment Terms (Days), and Credit Period (Days) — together significantly impact the dependent variable, Credit Utilization.

As per the first table, the F-value is 1.257, and the significance level (Sig.) is 0.2865, which is above 0.05. Consequently, the model is not statistically significant at the 5% level. In other words, the variables in the model do not jointly have a strong or meaningful impact on credit utilization. Hence, the study finds that changes in payment delay, credit period, or payment terms do not contribute significantly to explaining the variations in credit utilization.

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.112 ^a	.013	.003	8.968

a. Predictors: (Constant), Avg_Payment_Delay (Days), Delay (Days), Payment_Terms (Days), Credit_Period (Days)

(Table 4.2.14 H1 Model Summary)

INTERPRETATION

The R value indicated in the model summary is 0.112, which means that there is a very weak positive correlation between the independent variables — Average Payment Delay, Delay, Payment Terms, Credit Period — and the dependent variable, Credit Utilization. The R Square value is 0.013, indicating that the credit utilization variation accounted for by the predictors is only 1.3%. The rest 98.7% of the variation is due to other factors that have not been included in the model. The Adjusted R Square value of 0.003 also points to the fact that the model does not have sufficient explanatory power. The standard error of the estimate (8.968) is an indication that the model's predictions are associated with a relatively high level of error. Put simply, it implies that the variables chosen have very little impact on credit utilization and that there may be other factors that are more significant.

- **Objective 3: To identify any risks or delays caused by existing purchase and payment processes.**

Null Hypothesis (H_0): The terms of purchase in HAL's contracts have no significant impact on the efficiency of the procurement process.

Alternative Hypothesis (H_1): The terms of purchase in HAL's contracts have a significant impact on the efficiency of the procurement process.

Descriptive Statistics					
	N	Minimum	Maximum	Mean	Std. Deviation
Purchase_Value (Cr)	400	.02	14.86	2.4617	2.87605
Payment_Terms (Days)	400	30	90	52.42	18.333
Delivery_Period (Days)	400	22	304	151.82	56.354
Actual_Delivery (Days)	400	2	330	157.91	62.850
Delay (Days)	400	-80	120	6.10	29.344
Valid N (listwise)	400				

(Table 4.2.15 H2 Descriptive Statistics)

INTERPRETATION

The descriptive statistics detail HAL's purchase terms and procurement efficiency based on 400 records. The Contract values extend from ₹0.2 Cr to ₹14.86 Cr with an average of ₹2.46 Cr and the standard deviation of 2.88, it is showing a great deal of variation of contract sizes. The Payment Terms were limited to a range of 30 to 90 days, 52.42 days being the average with a standard deviation of 18.33, and this shows moderate differences in the credit period given to suppliers. The Delivery Period has a very broad range, from 22 to 304 days, averaging 151.82 days with a very high standard deviation of 56.35, and from this we can conclude that the delivery timelines differ a lot across contracts.

Actual Delivery has also a very wide range and in fact it varies from 2 to 330 days with an average of 157.91 days and a standard deviation of 62.85, this indicates that some suppliers deliver earlier or later than expected. The delay in delivery ranges from -80 to 120 days, with the average delivery delay being 6.10 days and a standard deviation of 29.34, and this reflects that while some deliveries are ahead of the schedule, there are others that are considerably late.

By and large, these statistics reveal that there is a considerable fluctuation in purchase terms and delivery performance, which provides the background to determine if purchase terms have an impact on procurement efficiency.

Correlations						
		Payment_Terms (Days)	Delivery_Period (Days)	Actual_Delivery (Days)	Delay (Days)	Purchase_Value (Cr)
Payment_Terms (Days)	Pearson Correlation	1	-.003	.021	.049	.003
	Sig. (2-tailed)		.959	.681	.327	.949
	N	400	400	400	400	400
Delivery_Period (Days)	Pearson Correlation	-.003	1	.884***	-.026	.025
	Sig. (2-tailed)	.959		<.001	.601	.613
	N	400	400	400	400	400
Actual_Delivery (Days)	Pearson Correlation	.021	.884***	1	.443***	-.020
	Sig. (2-tailed)	.681	<.001		<.001	.692
	N	400	400	400	400	400
Delay (Days)	Pearson Correlation	.049	-.026	.443***	1	-.091
	Sig. (2-tailed)	.327	.601	<.001		.068
	N	400	400	400	400	400
Purchase_Value (Cr)	Pearson Correlation	.003	.025	-.020	-.091	1
	Sig. (2-tailed)	.949	.613	.692	.068	
	N	400	400	400	400	400

***. Correlation at 0.001 (2-tailed)

(Table 4.2.16 H2 Correlations)

INTERPRETATION

The correlation analysis for Hypothesis 2 shows that most purchase terms have little or no significant impact on procurement efficiency. Out of 400 records, Payment Terms and Purchase Value are the two variables that have the least correlations with Delivery Period, Actual Delivery, and Delay, thus the length of credit or the size of the contract does not seem to affect procurement performance in any meaningful way.

The Delivery Period, on the other hand, is very strongly positively correlated with Actual Delivery ($r = 0.884$, $p < 0.001$), which means that longer planned delivery periods go together with longer actual delivery times. Also, Actual Delivery has a moderate positive correlation with Delay ($r = 0.443$, $p < 0.001$), thus delays are most of the time caused by the execution phase and not the contractual terms. In general, only the agreed Delivery Period has an effect on actual delivery and delays while other purchase terms, e.g., Payment Terms and Purchase Value, virtually have no impact on procurement efficiency, thus operational factors determine the timeliness of procurement to a greater extent.

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.056 ^a	.003	-.002	29.373
a. Predictors: (Constant), Delivery_Period (Days), Payment_Terms (Days)				

(Table 4.2.17 H2 Model Summary)

INTERPRETATION

The regression model summary tells us that the purchase terms—Payment Terms and Delivery Period—have almost no impact on procurement efficiency (measured through Delay). The correlation coefficient $R = 0.056$ is indicative of an extremely weak relationship between the independent variables and the dependent variable. The R^2 value of 0.003 signifies that just 0.3% of the variation in procurement efficiency is due to these purchase terms, whereas the adjusted R^2 of -0.002 suggests that the model does not become more accurate in its predictions when taking into account the number of predictors. The standard error of 29.373 is indicative of a significant amount of variability in the data that the model is unable to account for. In sum, the regression findings point to the fact that Payment Terms and Delivery Period, considered together, do not have a significant effect on procurement efficiency, thus, implying that other operational, logistical, or supplier-related factors might be the main drivers of timely procurement performance.

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1064.803	2	532.401	.617	.540 ^b
	Residual	342510.395	397	862.747		
	Total	343575.198	399			
a. Dependent Variable: Delay (Days)						
b. Predictors: (Constant), Delivery_Period (Days), Payment_Terms (Days)						

(Table 4.2.18 H2 ANOVA)

INTERPRETATION

The ANOVA analysis, the purchase terms of HAL, namely Delivery Period and Payment Terms, have no significant influence on the efficiency of procurement, which is gauged by delivery delays. The F-value of 0.617 with a significance of 0.540 shows that the model does not make a significant move towards predicting the delays beyond simply using the average. Put differently, changes in delivery timelines or payment terms hardly account for the differences in suppliers' timely delivery.

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	4.039	5.968		.677	.499
	Payment_Terms (Days)	.079	.080	.049	.980	.328
	Delivery_Period (Days)	-.014	.026	-.026	-.521	.603
a. Dependent Variable: Delay (Days)						

(Table 4.2.19 H2 Coefficients)

INTERPRETATION

The regression coefficients show how each of HAL's purchase terms individually impact procurement efficiency (Delay). The constant (intercept) of 4.039 is the delay that is expected when both Payment Terms and Delivery Period are zero. Payment Terms have a tiny positive coefficient ($B = 0.079$, $Beta = 0.049$, $p = 0.328$), which indicates that there is a slight association of longer payment terms with higher delays, however, this effect is not statistically significant. Delivery Period has a very small negative coefficient ($B = -0.014$, $Beta = -0.026$, $p = 0.603$), pointing to an almost non-existent inverse relationship with delay that is also not significant. The coefficients, in fact, point to the same conclusion that the two variables of Payment Terms and Delivery Period do not significantly cause changes in procurement efficiency, thus, the results from the model summary, ANOVA, and correlation analyses all agree.

CHAPTER – 05

SUMMARY OF FINDINGS, CONCLUSIONS, AND SUGGESTIONS

5.1 SUMMARY OF FINDINGS

- This research aimed to study how credit terms and purchasing conditions affected supplier performance and the efficiency of procurement at Hindustan Aeronautics Limited (HAL).
- The main goal was to figure out if behaviors of suppliers are significantly influenced by formal agreements such as credit period, discounts for early payment, interest on late payment, payment terms, delivery periods, and purchase values and therefore, the processes of procurement become more efficient or not.
- Knowing that supplier management and procurement efficiency are the lifelines of operational success in big organizations like HAL and that understanding of the role of contractual terms would be helpful in decision-making and process improvements, this research was carried out.
- The research made use of secondary data extracted from 400 supplier records at HAL which allowed for a robust quantitative analysis. Variables manipulated included Credit Utilization, Average Payment Delay, Actual Delivery, and Delay, while the variables to manipulate consisted of Credit Period, Early Payment Discount, Late Payment Interest, Payment Terms, Delivery Period, and Purchase Value.
- The descriptive statistics showed a large variation of both credit and purchase terms. An illustration of this would be that credit periods were ranging from 30 to 400 days, early payment discounts differed vastly, and late payment interest was only moderately varied among suppliers.
- In the same manner, purchase terms like Payment Terms and Delivery Periods showed great differences, and actual delivery of goods by suppliers varied highly, some going as far as delivering before the deadline whereas others significantly delayed. These

discoveries not only brought to light the varied nature of supplier arrangements but also prepared the ground for correlation and regression analyses to be conducted to find out their impact on performance results.

- For Hypothesis 2, which looked at how purchase terms affected procurement efficiency, descriptive statistics also showed high variations in Payment Terms, Delivery Periods, and Purchase Values.
- This means that although longer Delivery Periods imply longer actual delivery times, overall procurement efficiency is not significantly affected by the payment and delivery terms specified in the contracts.
- Factors related to operations such as supplier reliability, process efficiency, and execution practices seem to have a much greater influence on the results. These results indicate that HAL's suppliers might be more influenced by operational management practices and relationship management than by the specific financial or contractual terms given to them.
- Summing up, the research points out that HAL needs to shift its focus more towards the operational side of the business.
- This includes enhancing supplier monitoring systems, the usage of real-time performance dashboards, and the improvement of communication and coordination in the procurement departments. Besides that, suppliers can be motivated through relationship-building activities, recognition programs, and providing leadership and skill-building opportunities that will make them more willing to comply and be timely.

5.2 CONCLUSION

- This research aimed at understanding how the terms of credit and purchase affect supplier performance and procurement efficiency at Hindustan Aeronautics Limited (HAL) in particular, through the exploration of the possibility of contractual arrangements as a tool to lead supplier behavior and procurement on time.
- The examination of 400 supplier records enabled the study to involve a great number of variables such as Credit Period, Early Payment Discount, Late Payment Interest, Payment Terms, Delivery Period, and Purchase Value; their interrelation with performance indicators like Credit Utilization, Average Payment Delay, Actual Delivery, and overall Delay of procurement were also contemplated.
- The research evidenced that credit facilities have almost no significant influence on the behavior of suppliers at HAL. The results of the correlation analysis revealed that the relationships between credit terms and supplier performance measures were weak. A somewhat negative relationship between credit period and early payment discount was reported, implying that longer credit periods usually bring about lower discounts offered by suppliers.
- Payment terms together with purchase value had no significant effect on procurement efficiency, thus, financial arrangements alone cannot guarantee better supplier compliance or on-time delivery.
- Nevertheless, the delivery period was confirmed to be the main factor of actual delivery performance which means that when HAL not only specifies but also monitors shorter delivery windows, the suppliers make sure to deliver as per the requirement.
- Such a change in focus from monetary aspects to non-financial factors such as dependability, process execution, coordination, and timely communication was a major finding in the study.

- Suppliers must get involved in the relationship-building activities rather than just be informed through communication and engagement since they in turn are the most loyal to these practices rather than financial penalties or interest clauses. The findings underline the importance of operational reliability, the capability to track in real-time, and building collaborative relationships with suppliers to achieve sustained efficiency.
- The results also highlight that nothing on its own, including finance-related incentives and penalties, can bring about a continuous performance of suppliers. The company may also create a supplier scorecard system that rewards those suppliers who are not only on time but also deliver quality products thereby healthy competition among vendors will be nurtured.
- Sustainability became a major pillar in the procurement strategy of HAL for the future. Moreover, the firm is constantly giving environmental responsibility a priority in her operations.
- The list of activities for this includes achieving carbon neutrality, raising the use of renewable sources of energy, lowering waste, and encouraging the vendors to use green supply chain practices. HAL can still keep its operations economically feasible and environmentally friendly by merging the sustainability goals with procurement efficiency. Besides, getting the employees more involved and the community participating will make it easier for HAL to internalize sustainability as one of the core values of her business.
- In brief, the research comes to the conclusion that credit and purchase terms only have minor impacts that are direct on supplier performance as well as procurement efficiency at HAL. The firm's ultimate success is thus grounded in operational excellence, digital transformation, and long-term collaboration with the suppliers.
- To bring about procurement reliability, HAL should keep on working on their supplier monitoring techniques, financial procedures streamlining, and communication channels strengthening across departments. The future strategy of HAL should be based on the use of predictive analytics, process automation, and performance-based supplier management.

- HAL will be able to maintain her competitive advantage in the defense sector by properly aligning her procurement methods with the global standards of transparency, sustainability, and innovation.
- The combination of financial discipline, operational agility, and sustainable supplier partnerships will create a win-win situation for HAL and thus she will be able to extend her role as a responsible, efficient, and forward-looking aerospace organization.
- Along with continuous strive on procurement efficiency and strategic supplier management, HAL will keep being a major contributor to India's defense production ecosystem and national vision of Atmanirbhar Bharat in an effective way. Moreover, the study highlights that HAL should consistently refine its internal operations as well as the coordination between different departments.
- To procure, pay the suppliers, and dispatch the goods in a smooth manner the departments of procurement, finance, and operations have to work as a concerted unit.
- The enhanced collaboration will do away with the redundant steps that have been there and, in addition, it will promote a culture of accountability.

5.3 SUGGESTIONS TO THE ORGANIZATION

- The analysis has been done on influence of the credit and purchase terms on the supplier performance and procurement efficiency at Hindustan Aeronautics Limited (HAL).
- This study is based on the examination of 400 supplier records with a focus on variables such as Credit Period, Early Payment Discount, Late Payment Interest, Payment Terms, Delivery Period, and Purchase Value, and their connection with indicators of performance including Credit Utilization, Average Payment Delay, Actual Delivery, and overall Delay.
- The descriptive statistics have shown a wide range of differences in credit and purchase terms that point to significant variances in supplier arrangements and procurement practices.
- These results point to the fact that the operational side, supplier reliability, and process management matter more than the contractual terms for the procurement results.
- Reflecting on such outcomes, the paper arrives at a conclusion that the suppliers of HAL are more responsive to the operational and managerial practices rather than the formal terms of the contracts. Credit and purchase terms may have a little effect, but they are not enough to bring about substantial improvements in supplier performance or procurement efficiency.
- Firstly, HAL is recommended to enhance supplier monitoring mechanisms by the use of real-time dashboards and performance tracking, accelerating the procurement processes so as to lessen bottlenecks, and endorsing workflow automation as a way of fostering operational efficiency.
- Making supplier engagement robust through continuous feedback, recognition programs, and performance-based incentives can elevate the level of accountability and thus, the suppliers will be encouraged to deliver the required quantity and quality standards.

- Also, personalized credit and purchase terms for strategic suppliers along with predictive analytics may serve as a double agent to help decision-making be proactive and interventions timely.
- Moreover, procurement staff can be more efficient if they are equipped with the skills of negotiation, supplier evaluation, and process management, coupled with structured feedback mechanisms, and adopting supplier segmentation based on criticality and past performance.
- All these will lead to resources being efficiently allocated and ensuring that the high-priority suppliers get the most attention.
- By implementing these strategies, HAL will be able to increase procurement efficiency, lessen the occurrence of delays, improve supplier performance, and thus, the overall organizational effectiveness will be at a higher level.
- In this way, they will be able to transcend from a merely contract-driven approach to a comprehensive, strategic, and operationally robust procurement system.

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